

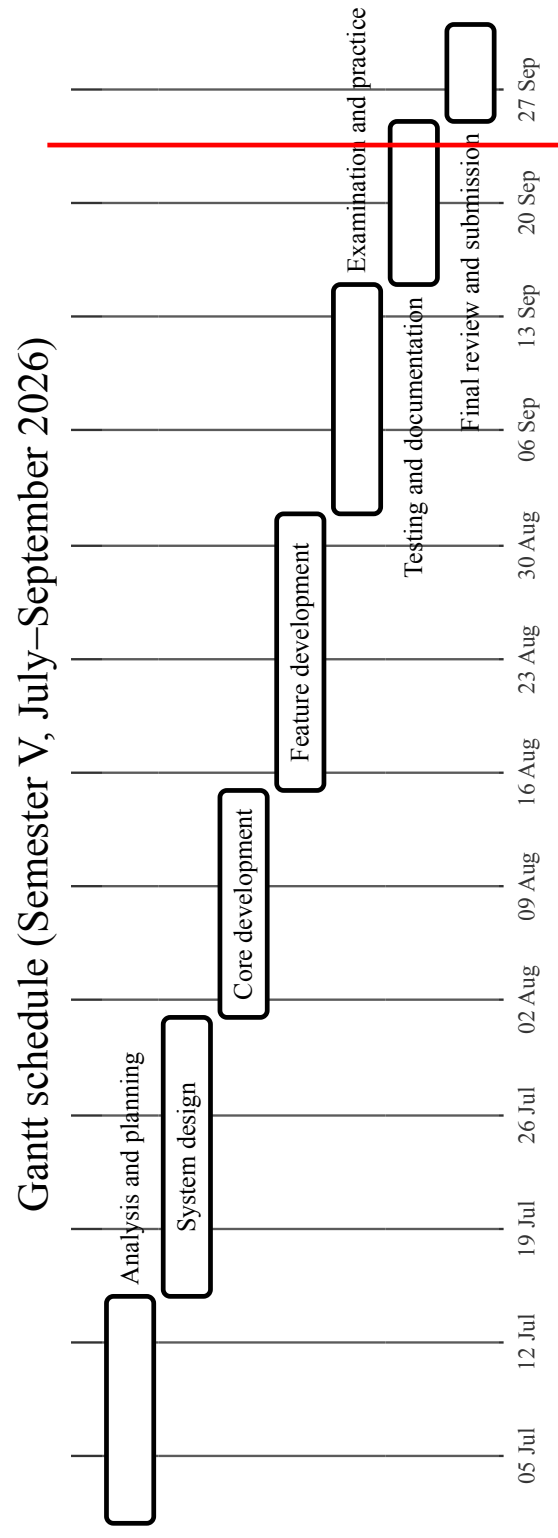
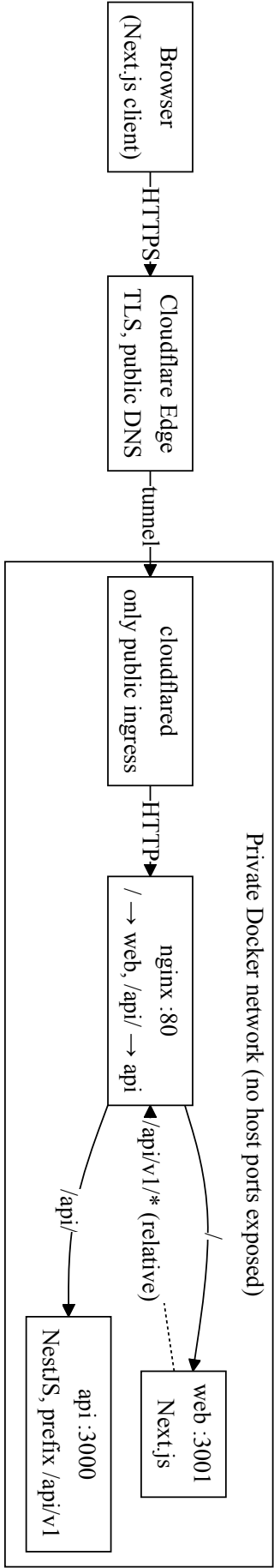


Figure 4.1. Gantt chart of the seven project phases (July–September 2026).

Catlum EduTech — System Architecture: Public Ingress and Routing



Catlum EduTech — System Architecture: Internal Services

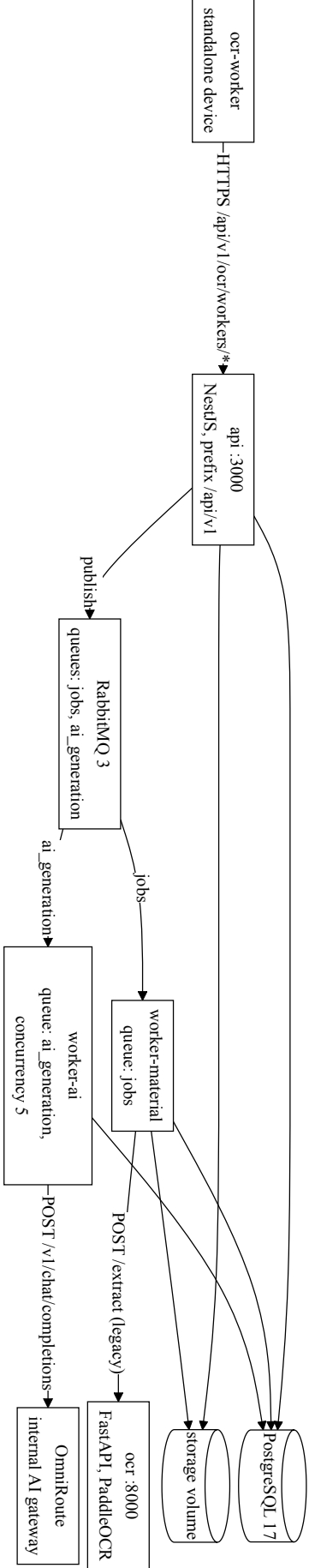


Figure 5.1. System architecture and deployment topology: (left) the single public ingress through the Cloudflare Tunnel and the internal nginx routing; (right) the internal services on the private Docker network and their data flows.

CatLium EduTech — Level-0 Data Flow Diagram

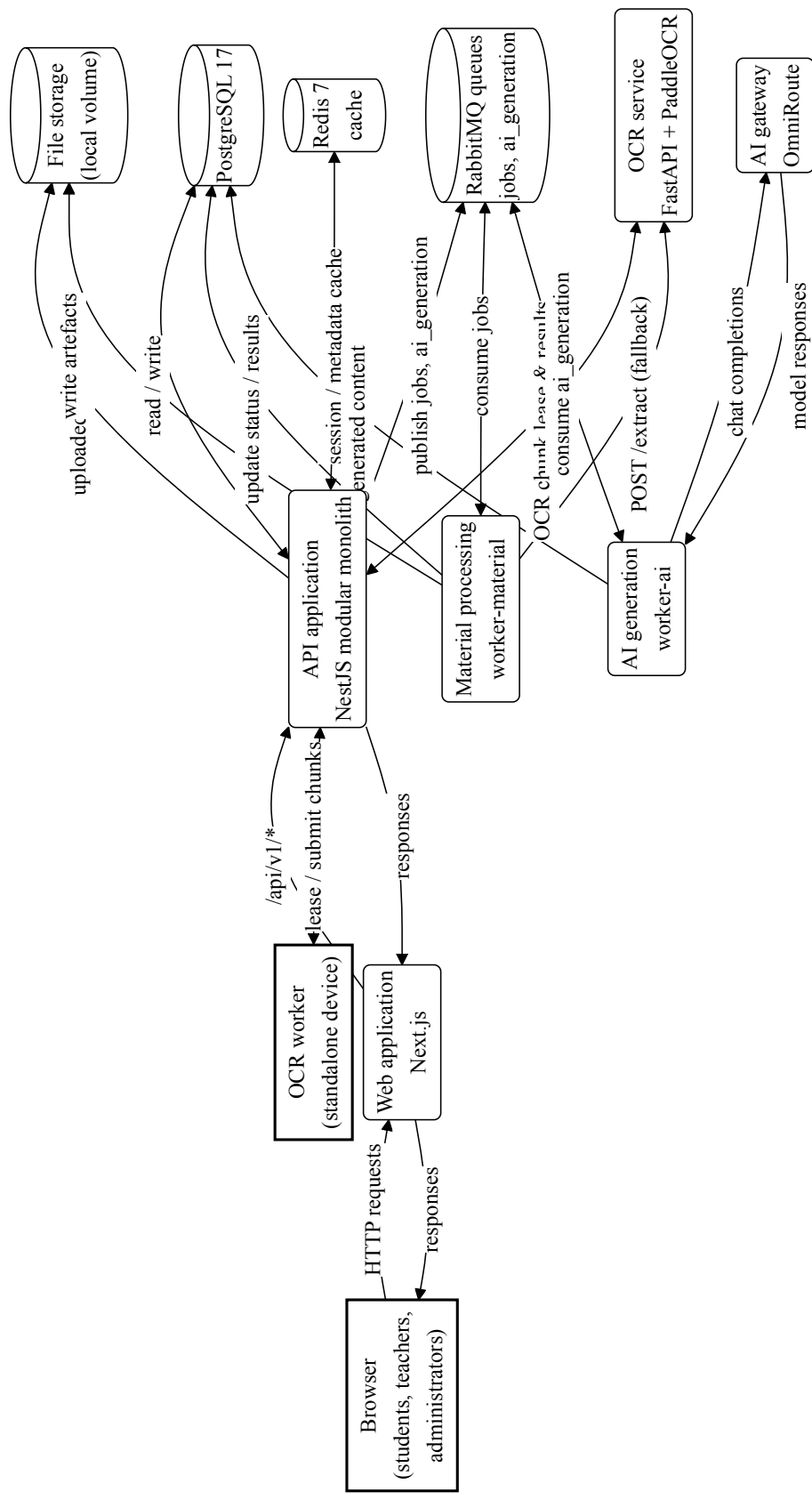


Figure 5.2. Level-0 data flow diagram of the system.

Call ium EduTech — Level-1 Data Flow Diagram (API decomposition)

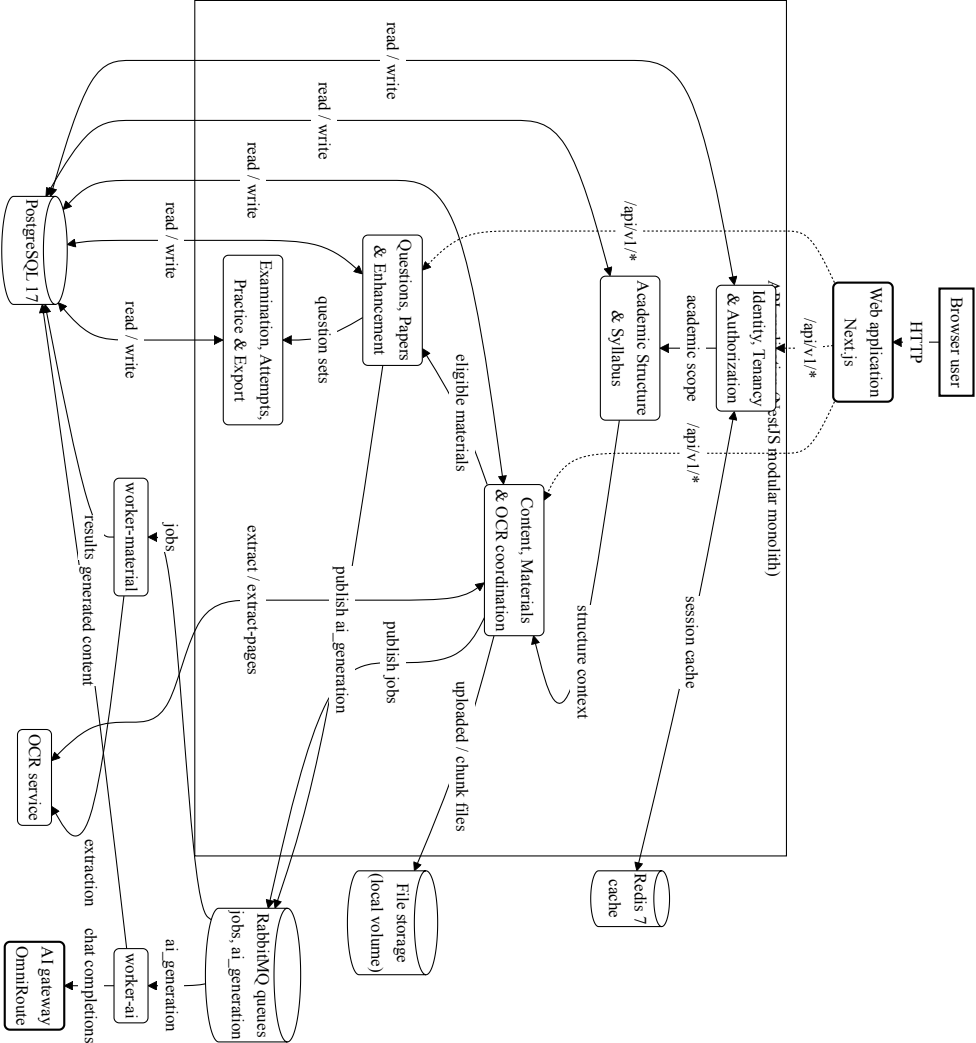


Figure 5.3. Level-1 data flow diagram: decomposition of the API application into its principal functional areas.

CatLium EduTech — Level-2 DFD (a): API-side coordination of OCR

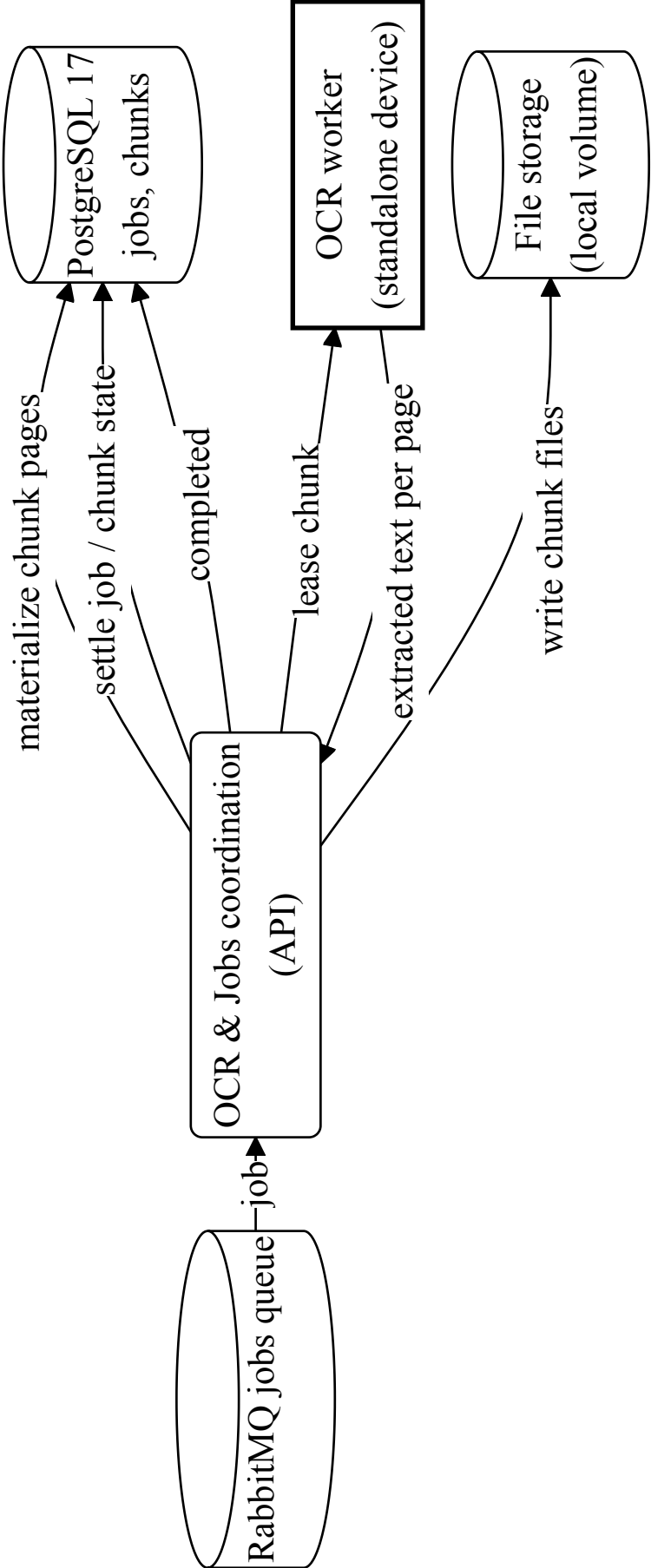


Figure 5.4. Level-2 data flow diagram — OCR and material-processing pipeline, part 1: API-side coordination of chunk leasing, aggregation, and job settlement.

Caillium EduTech — Level-2 DFD (b): OCR service extraction pipeline

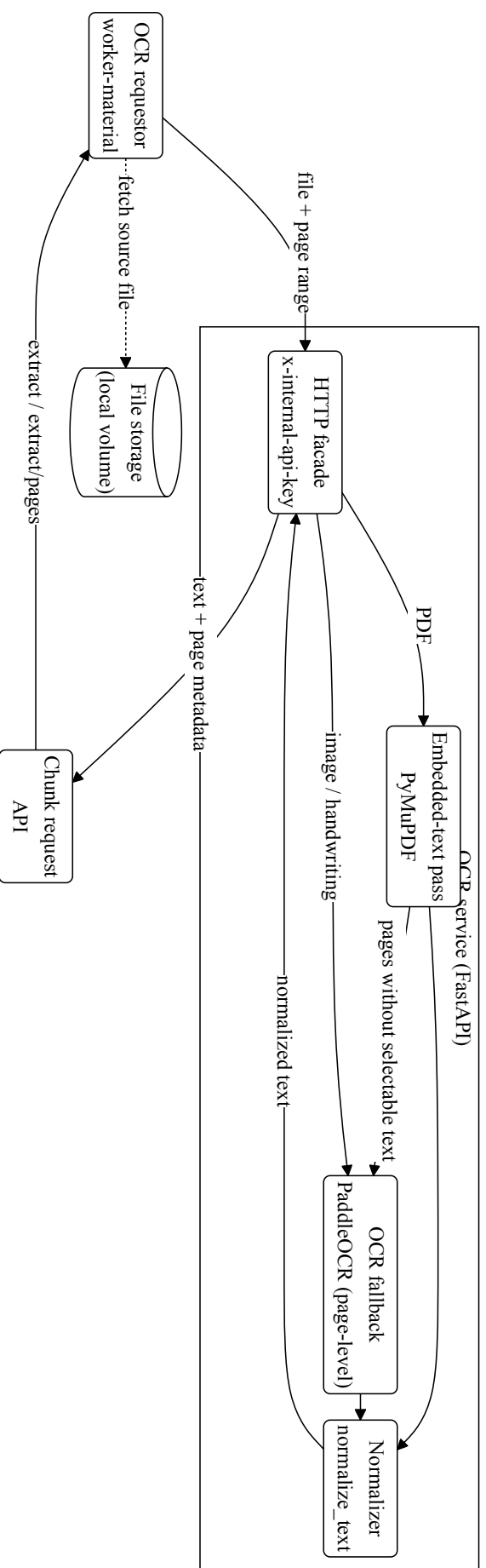


Figure 5.5. Level-2 data flow diagram — OCR and material-processing pipeline, part 2: the OCR service extraction pipeline (embedded text, PaddleOCR fallback, normalization).

CatLium EduTech — Level-3 Data Flow Diagram (AI generation pipeline)

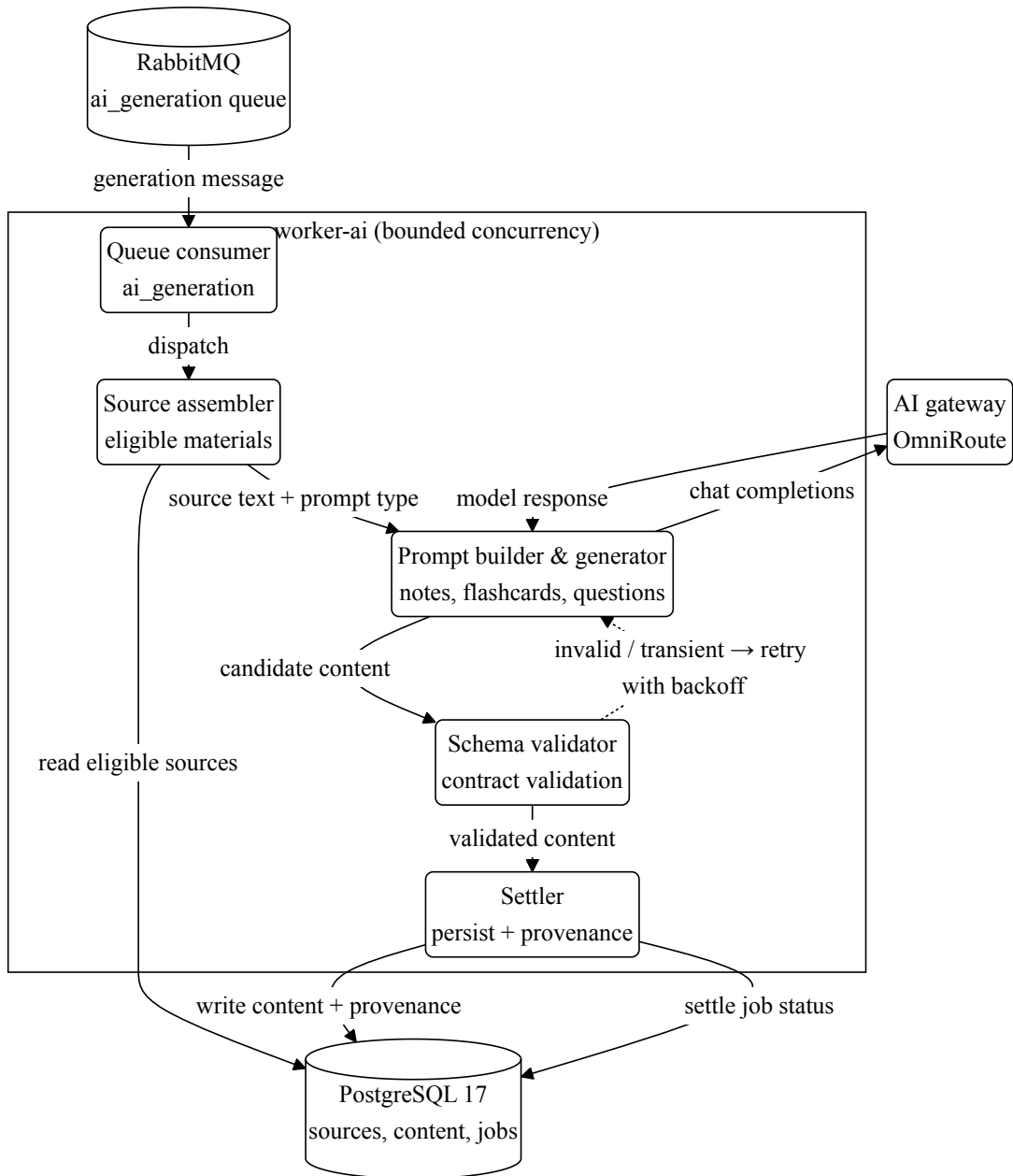


Figure 5.6. Level-3 data flow diagram: the AI generation pipeline inside `worker-ai`.

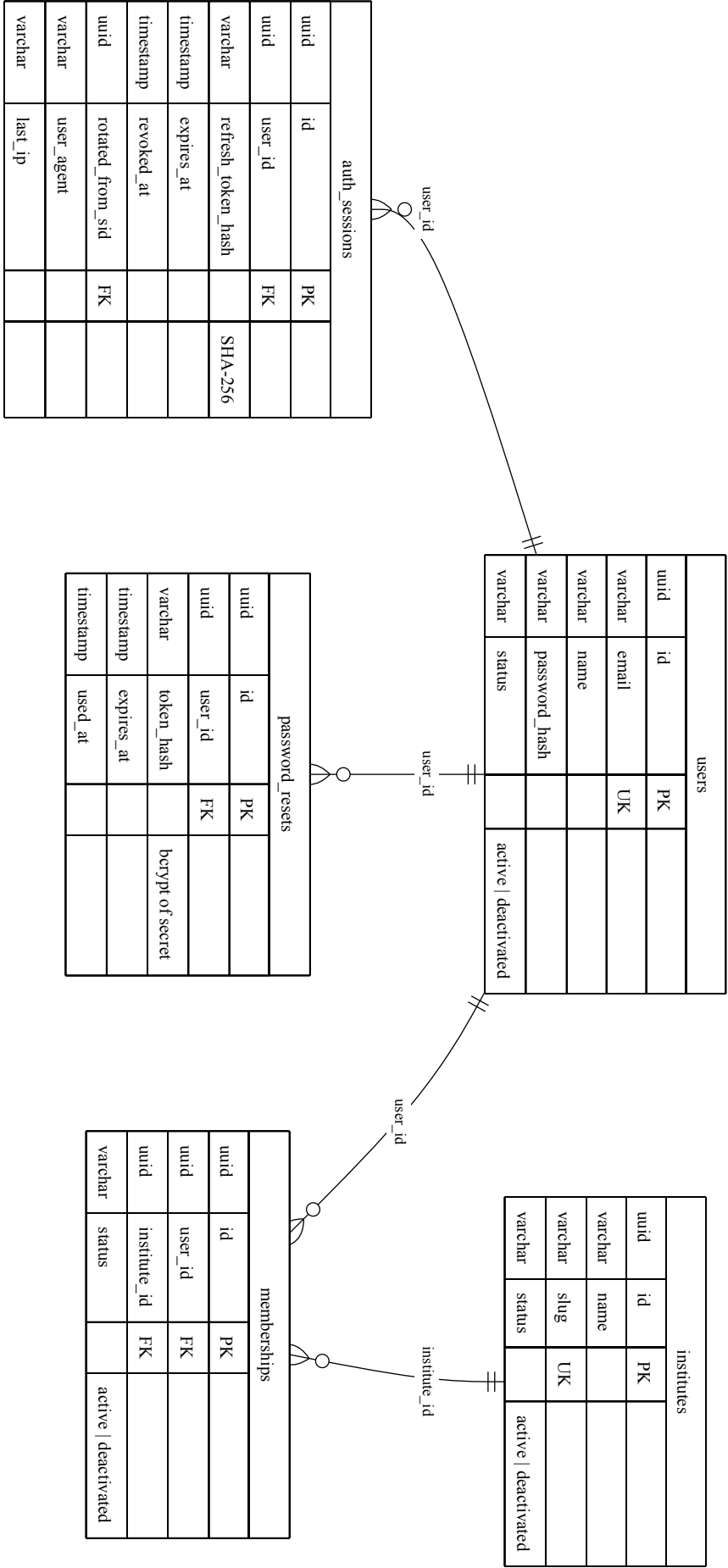


Figure 6.1. Database schema — the identity and authentication core: users, institute memberships, sessions, and password resets.

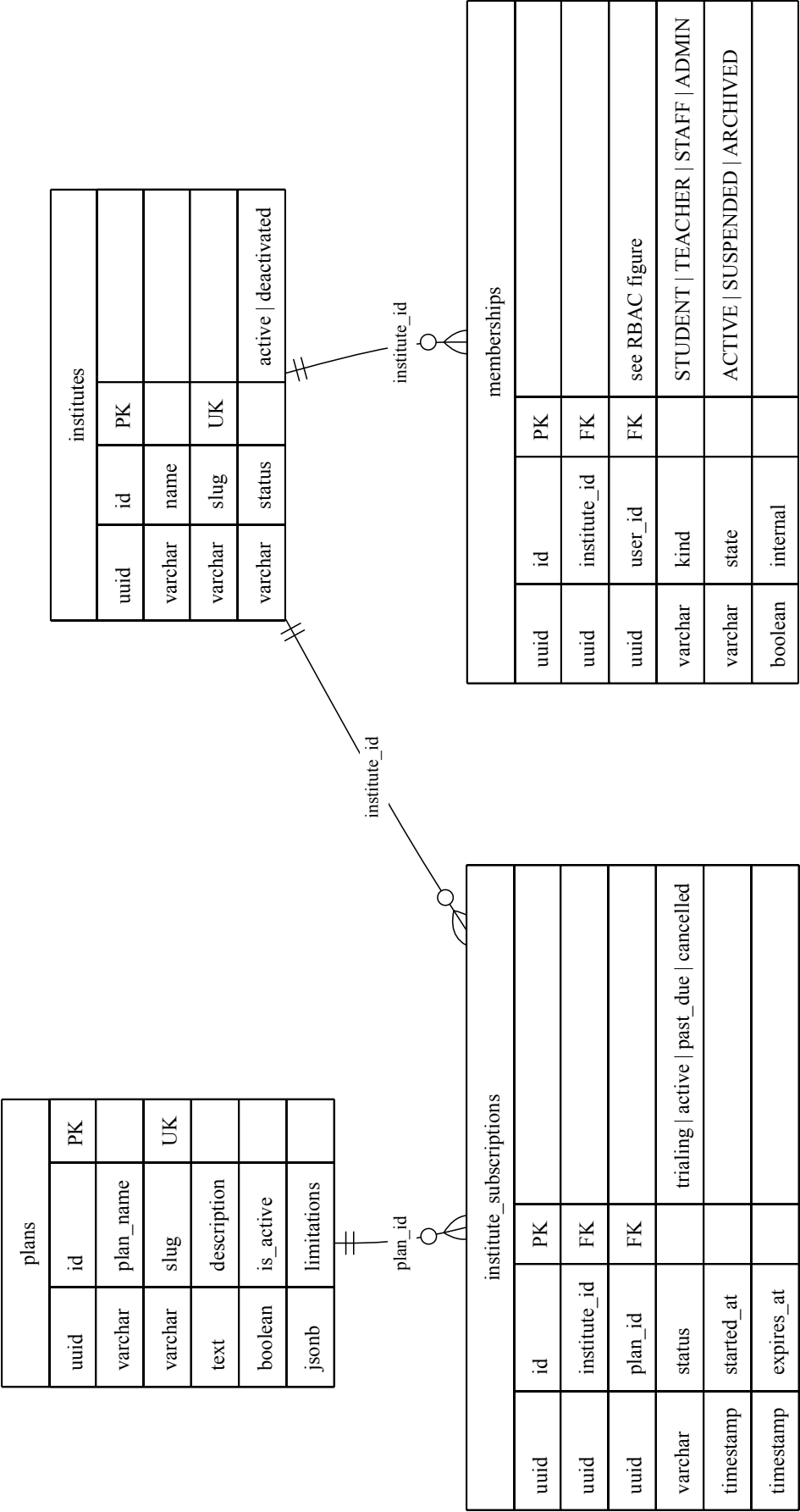


Figure 6.2. Database schema — the tenancy core: institutes, memberships, and subscription plans.

Carl ium EduTech — Database Schema (1c): Roles and Permissions (RBAC)

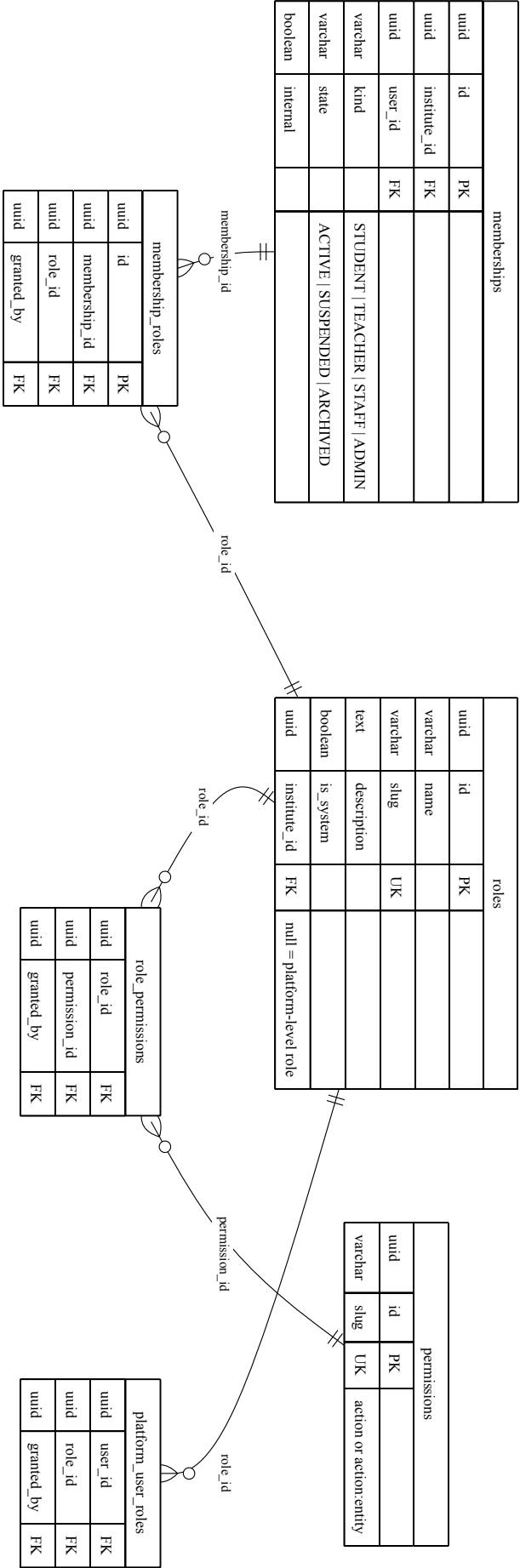


Figure 6.3. Database schema — the roles/permissions (RBAC) core.

CatLium EduTech — Database Schema (2a): Academic Structure and Offerings

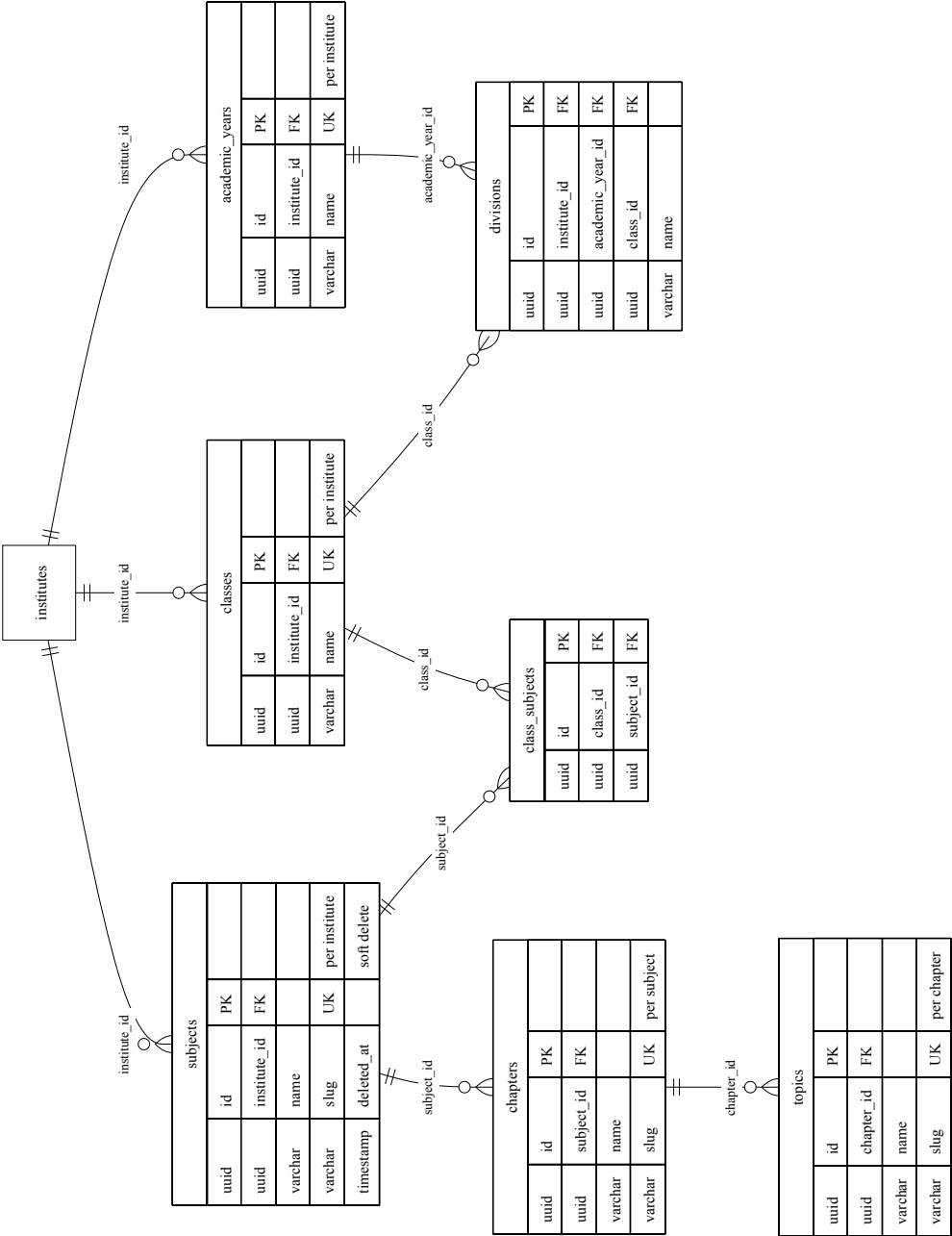


Figure 6.4. Database schema — academic structure: the subject chain (`subjects` → `chapters` → `topics`), class offerings, and academic years.

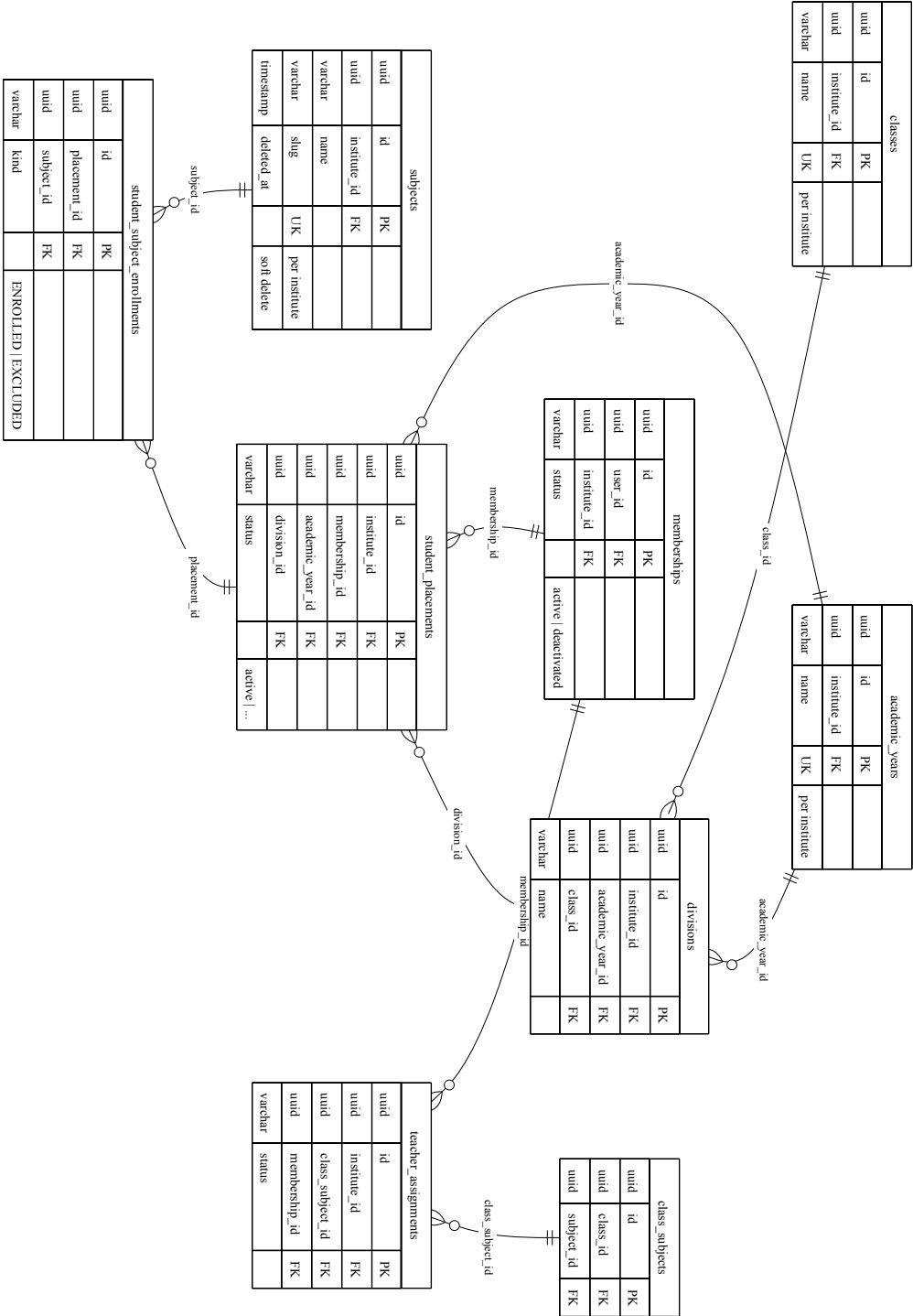


Figure 6.5. Database schema — placements and assignments that derive the academic scope for students and teachers.

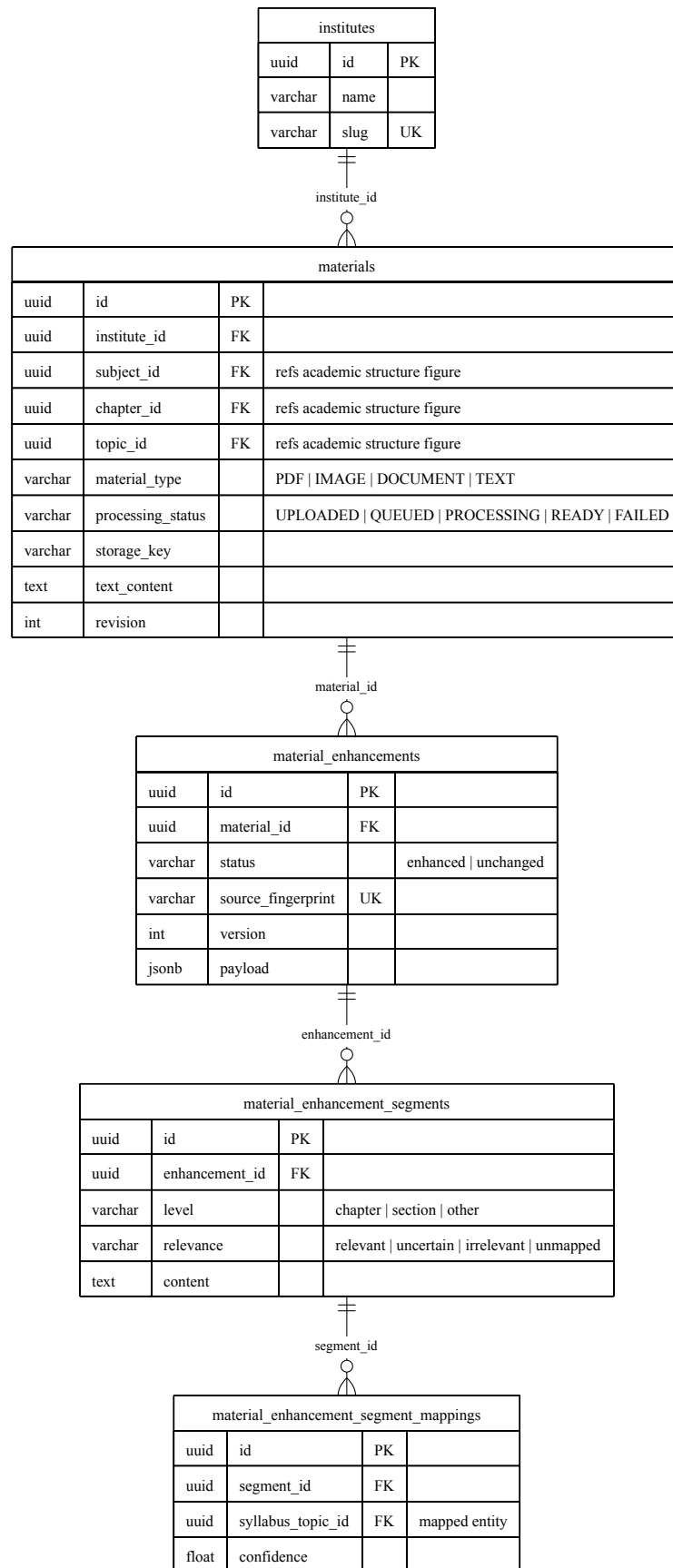


Figure 6.6. Database schema — materials with their deterministic enhancement chain.

CatLium EduTech — Database Schema (2d): Content Items and Versions

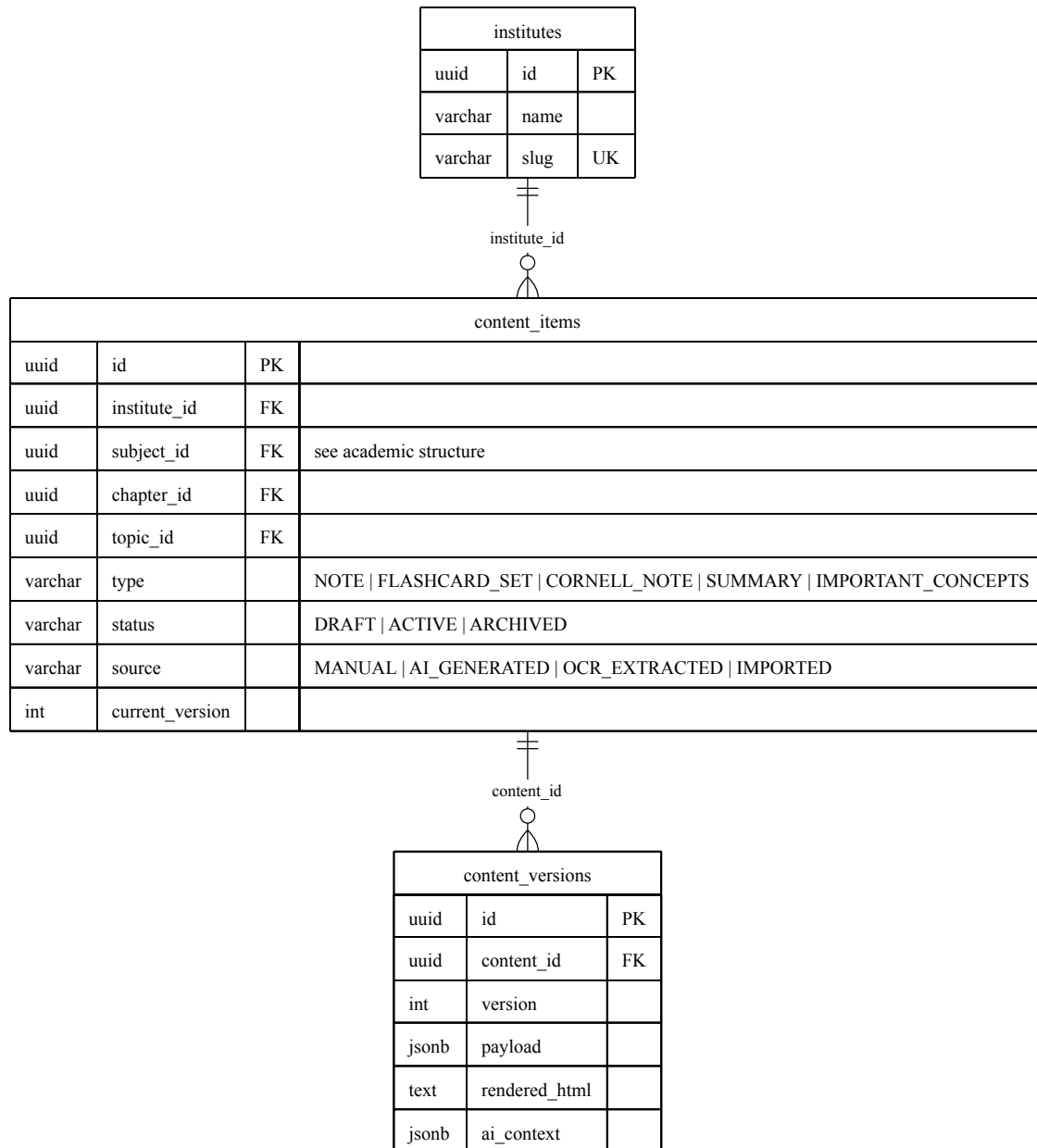


Figure 6.7. Database schema — versioned content items.

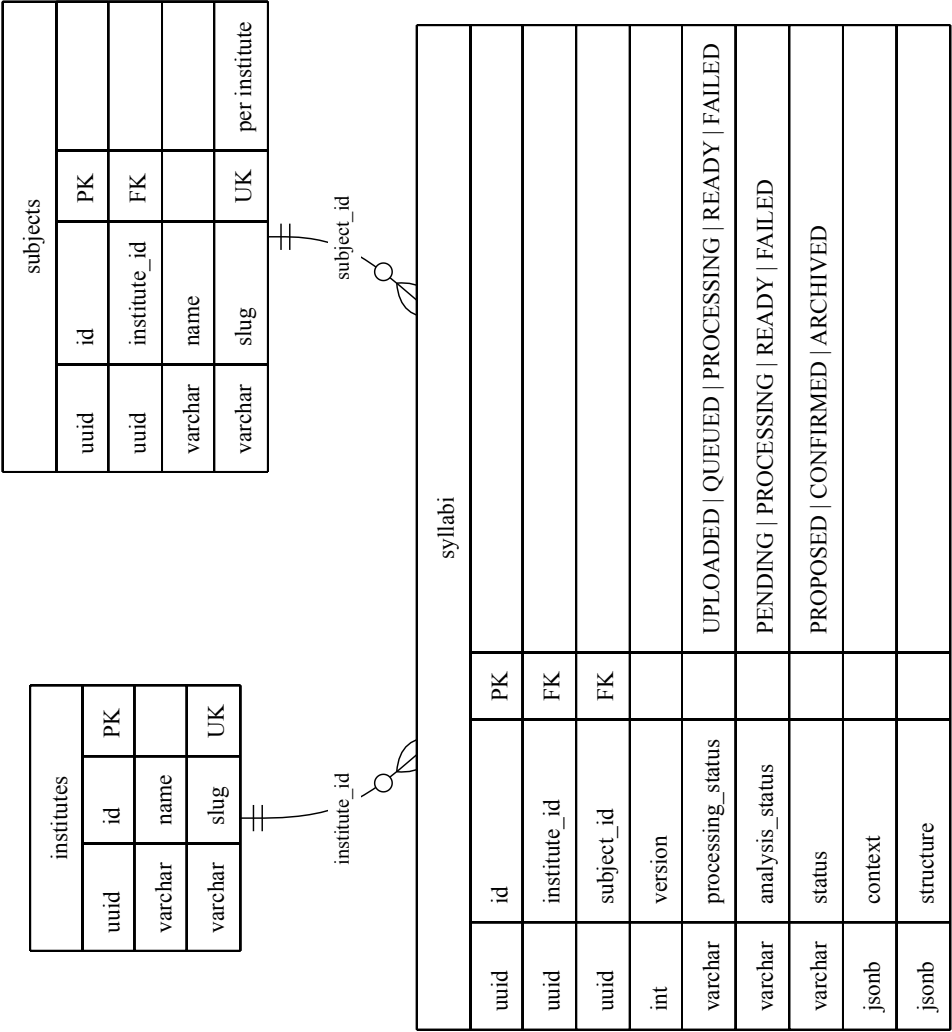


Figure 6.8. Database schema — per-subject syllabi and their lifecycle.

CatLium EduTech — Database Schema (3a): Question Bank

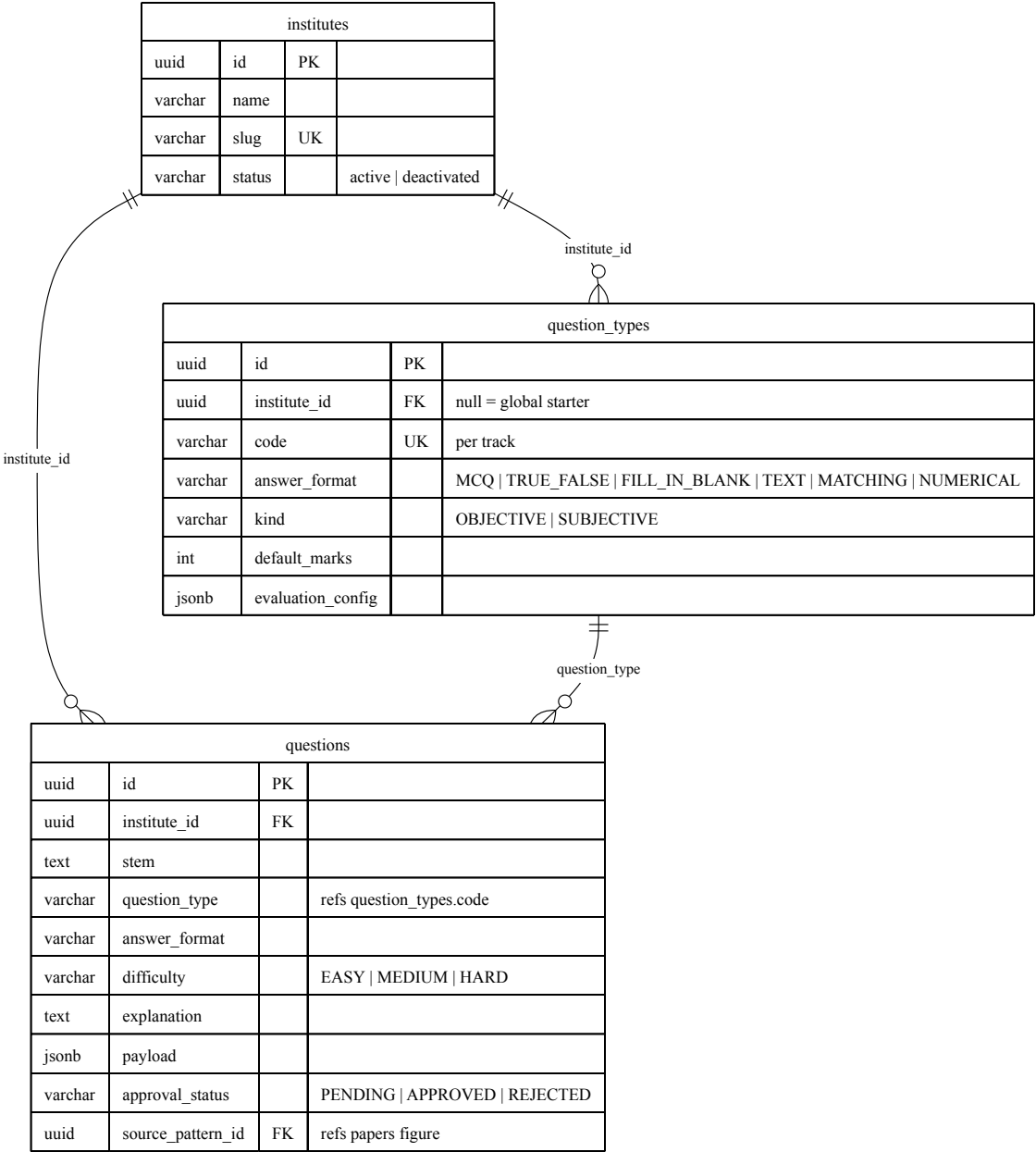


Figure 6.9. Database schema — question bank: question types and the scoped `questions` table.

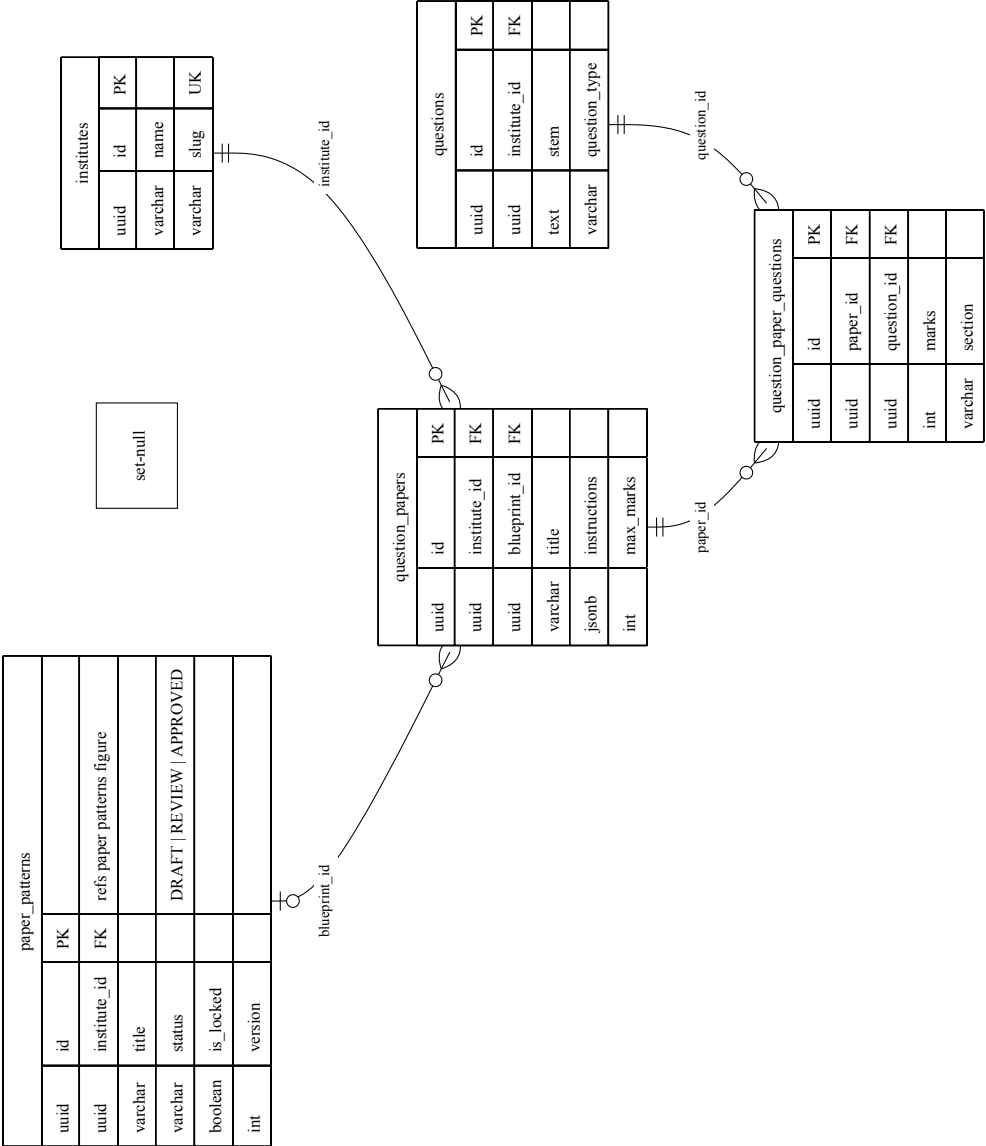


Figure 6.10. Database schema — question papers: the blueprint-driven question_papers snapshot.

Carl Jann EduTech — Database Schema (3c): Assessments and Attempts

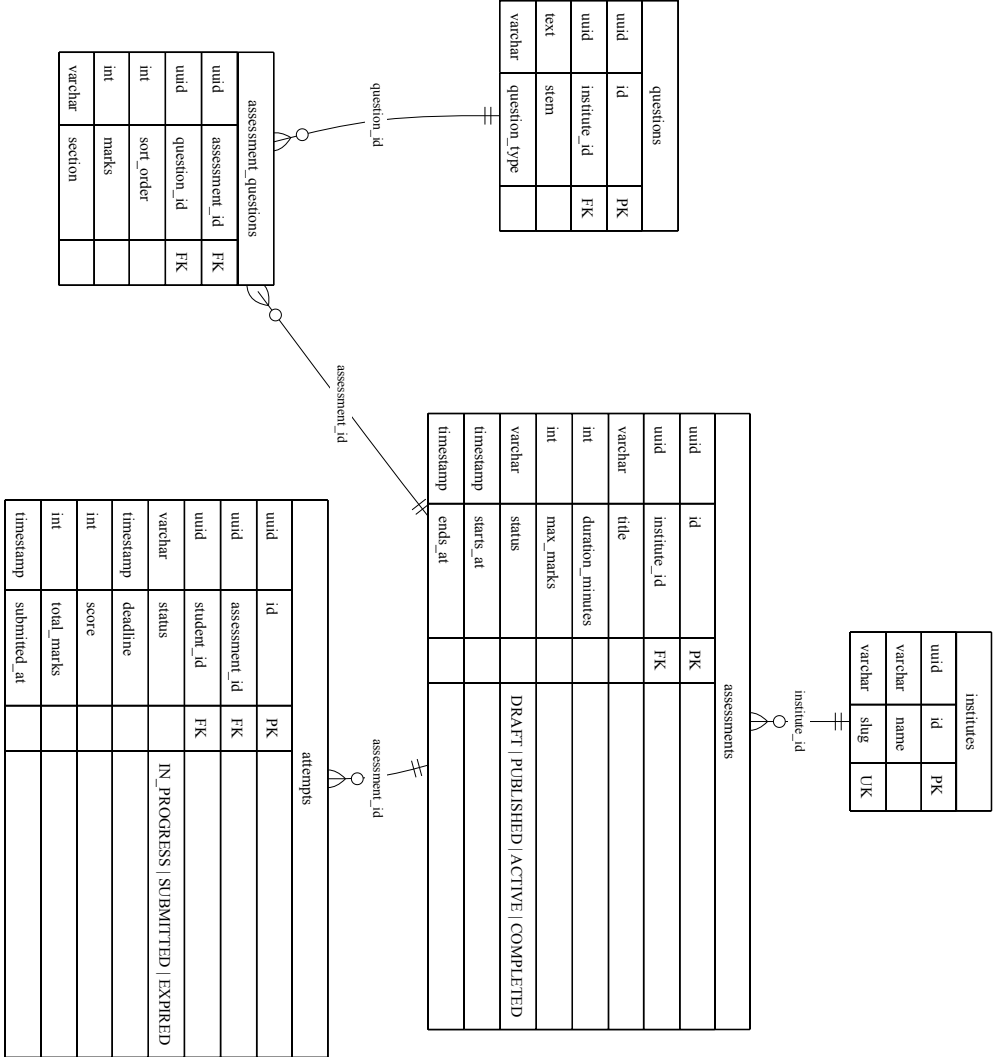


Figure 6.11. Database schema — examination: assessments and their subject binding.

CatLium EduTech — Database Schema (3d): Attempt Responses

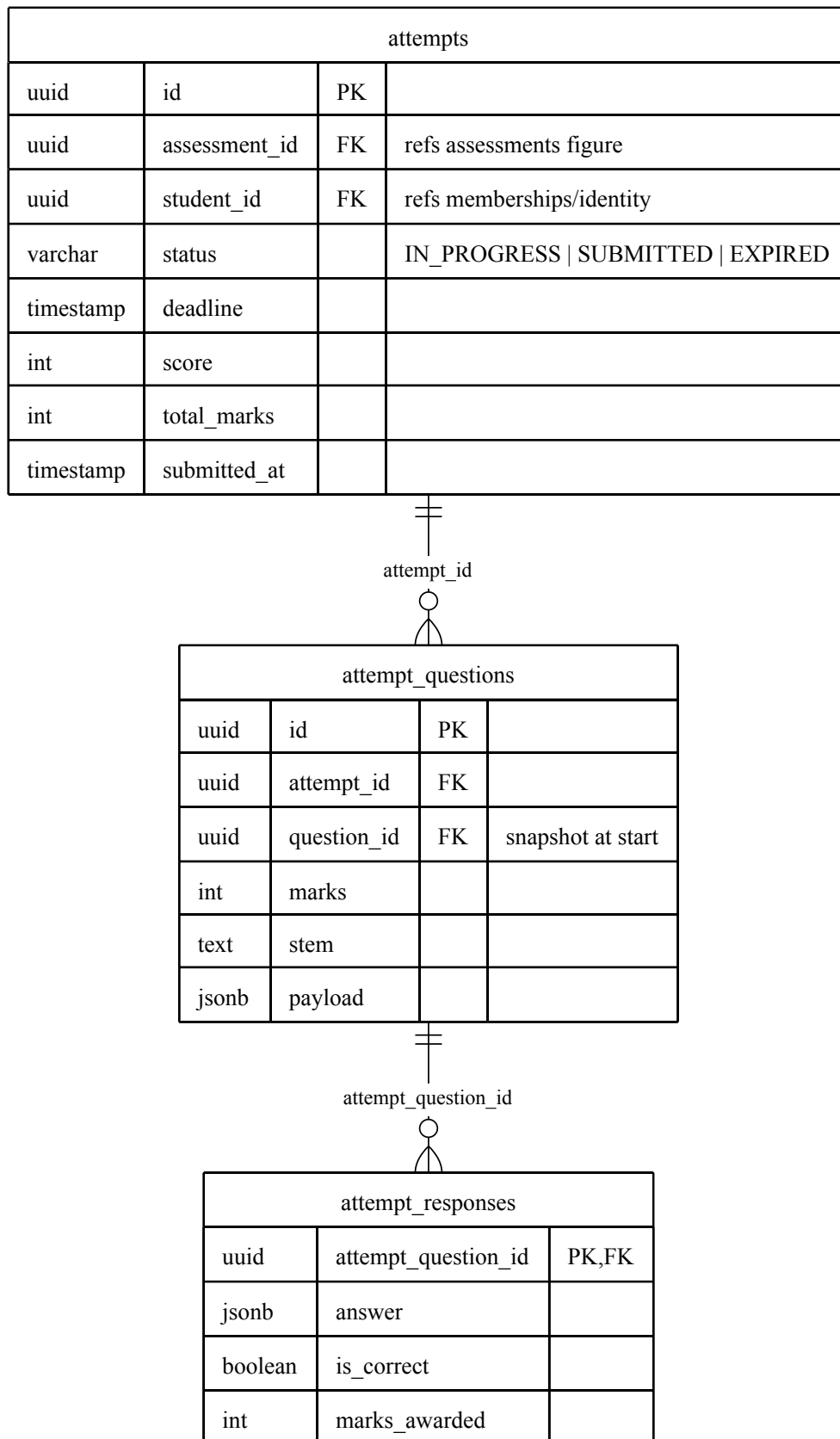


Figure 6.12. Database schema — attempt snapshots and response tables.

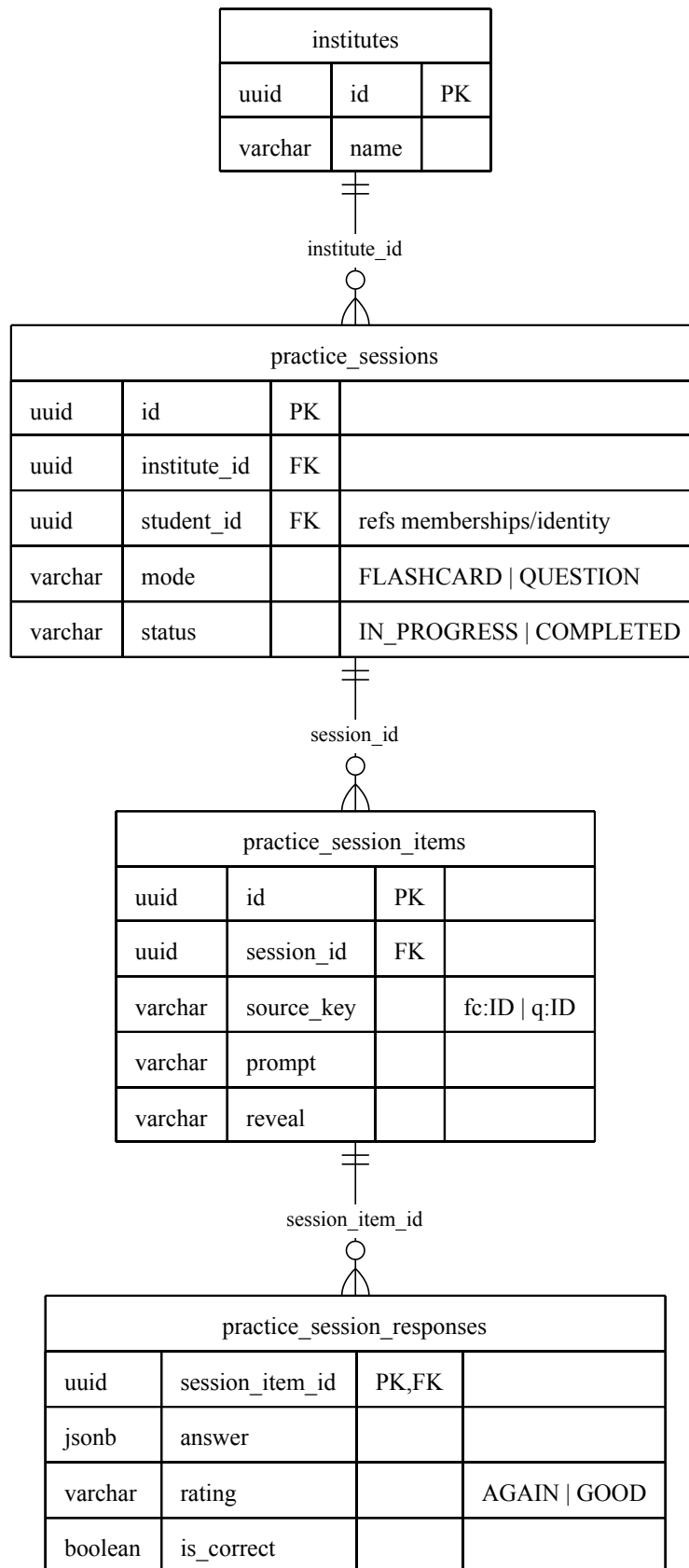


Figure 6.13. Database schema — ungraded practice sessions.

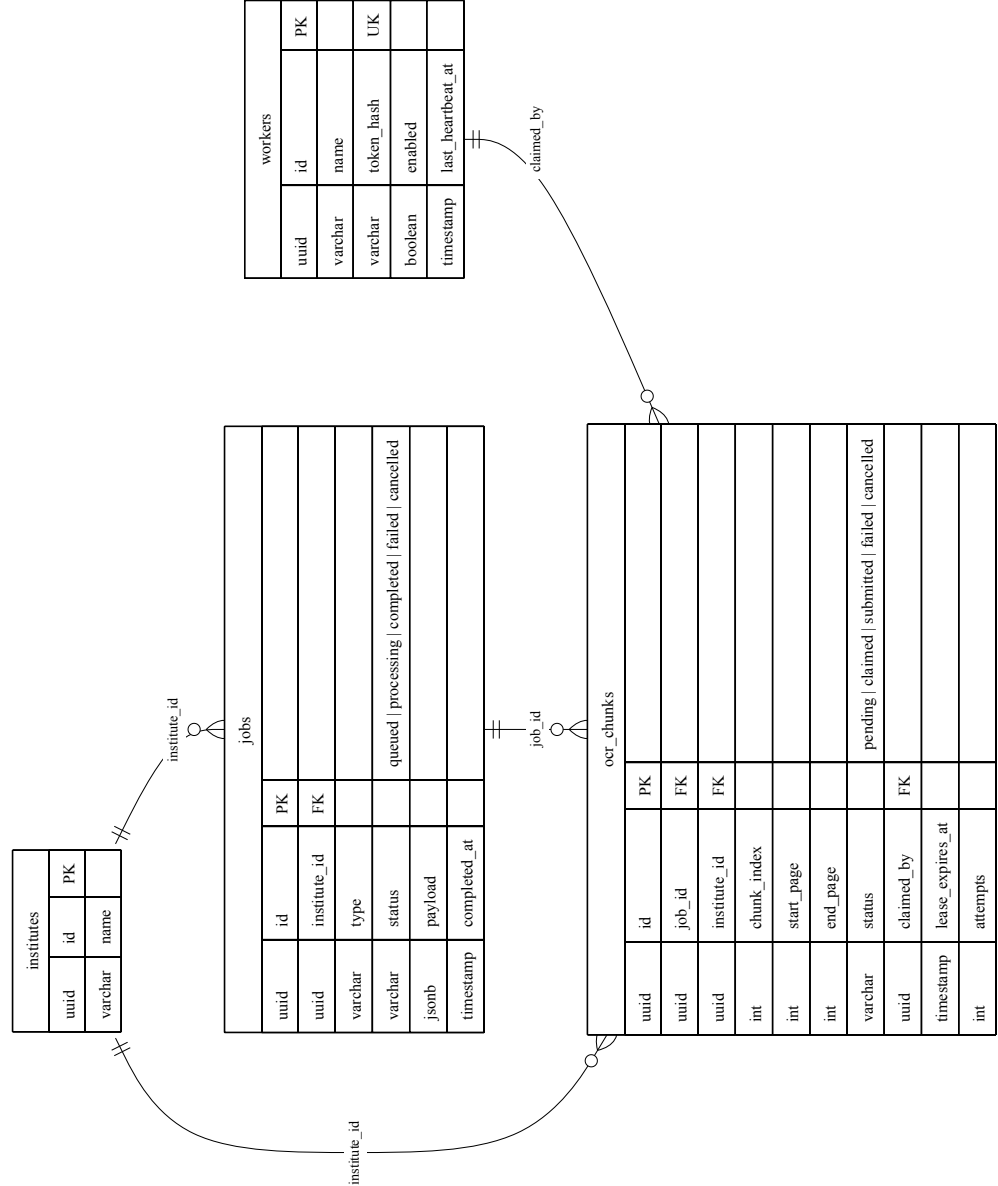


Figure 6.14. Database schema — operations: the job/worker/OCR-chunk registry.

Carl Lum Edu Tech — Database Schema (3b): Paper Pattern Blueprints

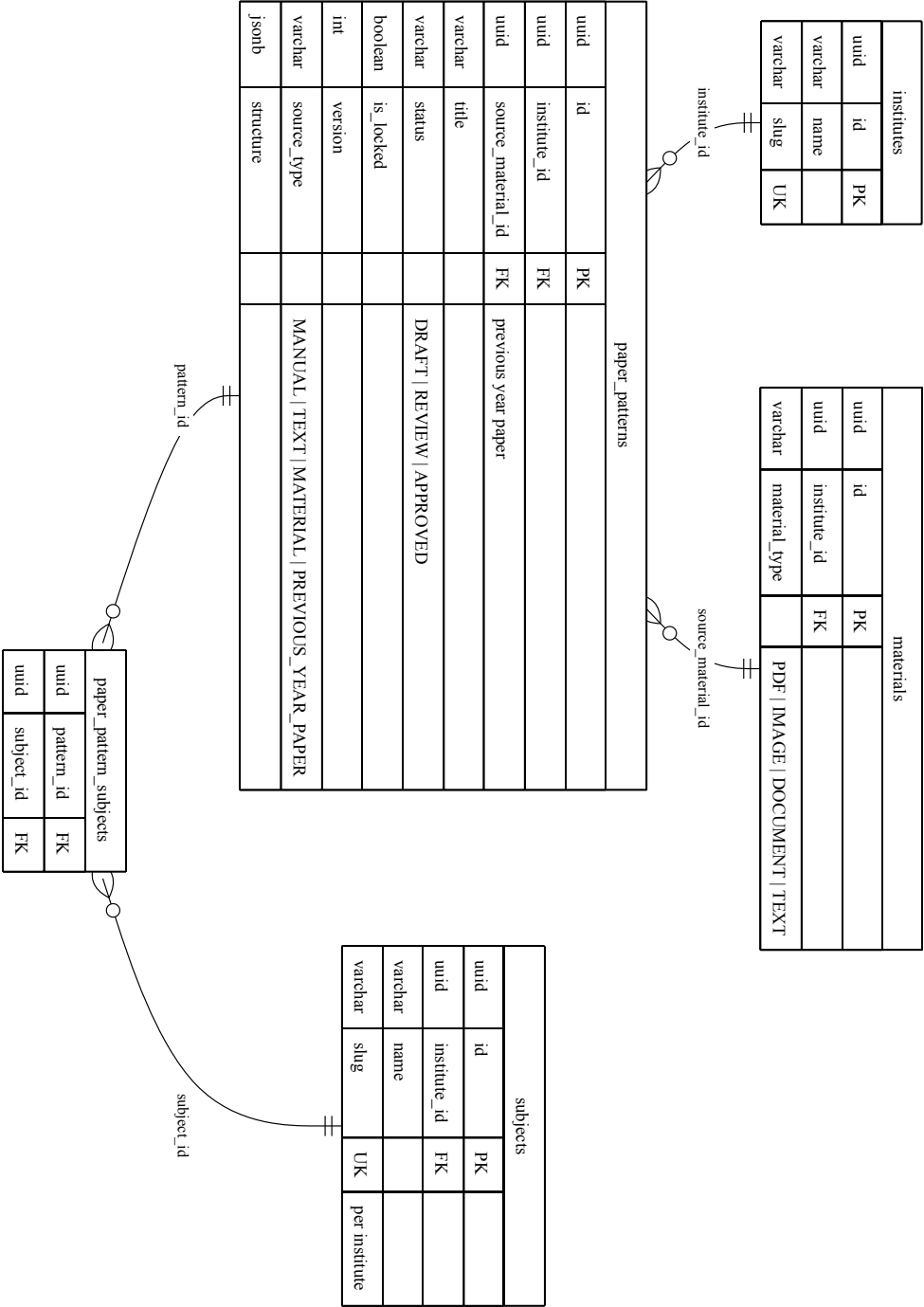


Figure 6.15. Database schema — operations: paper-pattern blueprints.

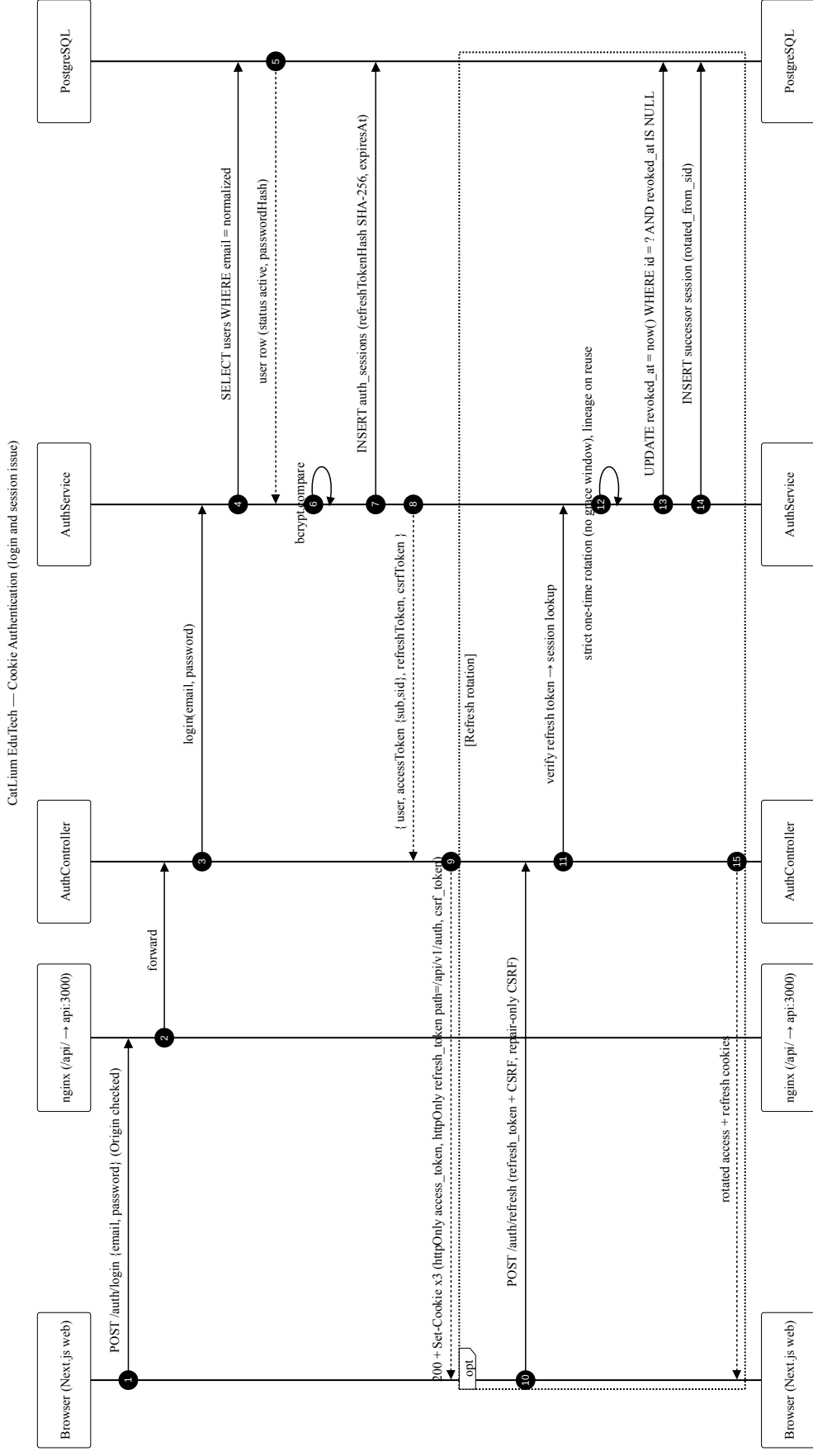


Figure 7.1. Cookie authentication: login, cookie issue, and strict one-time refresh rotation.

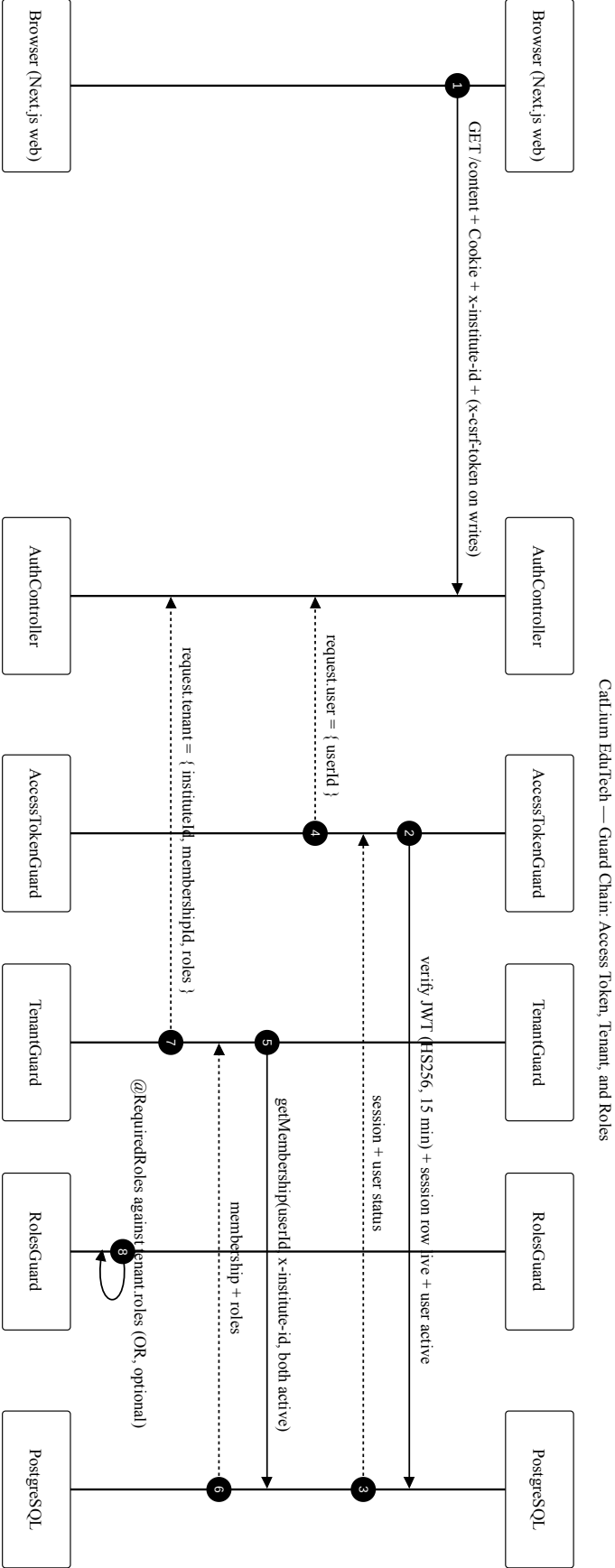


Figure 7.2. Tenant-scoped request flow, part 1: the access-token, tenant, and roles guards.

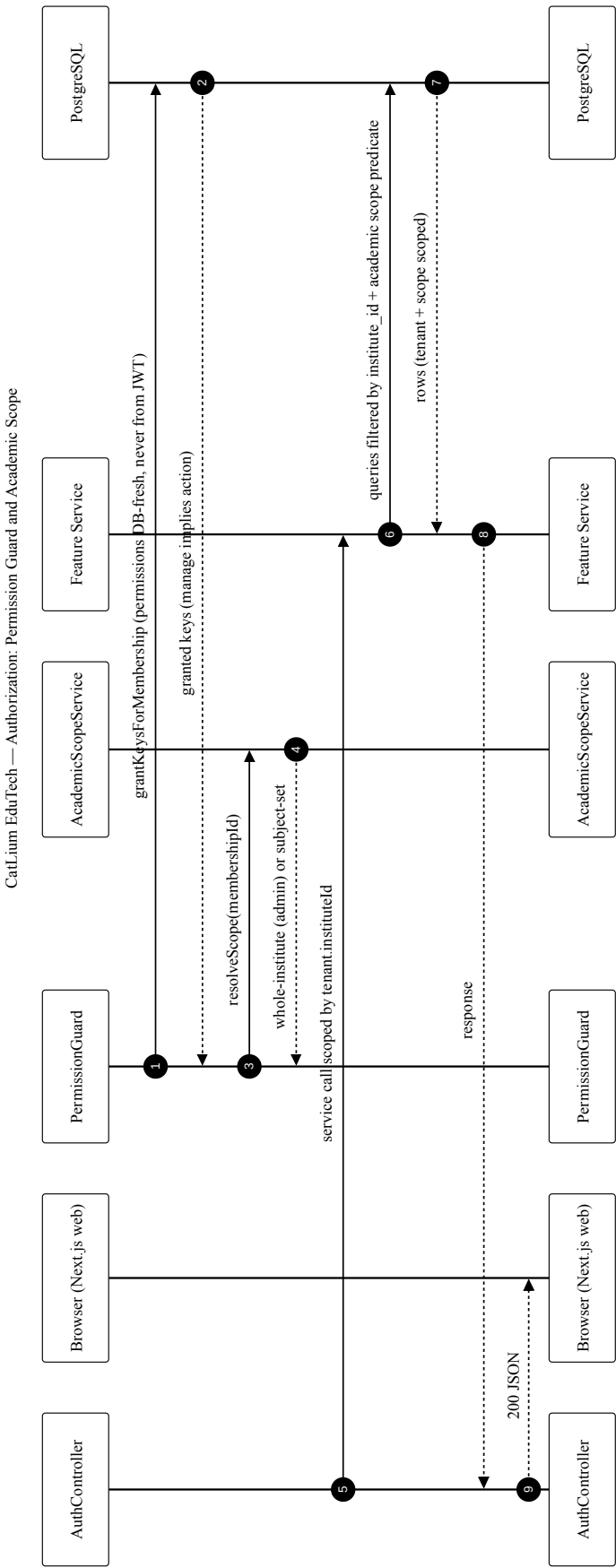


Figure 7.3. Tenant-scoped request flow, part 2: permission-guard evaluation and academic-scope enforcement in the feature service.

CatLium EduTech — Authorization Decision Chain (guards)

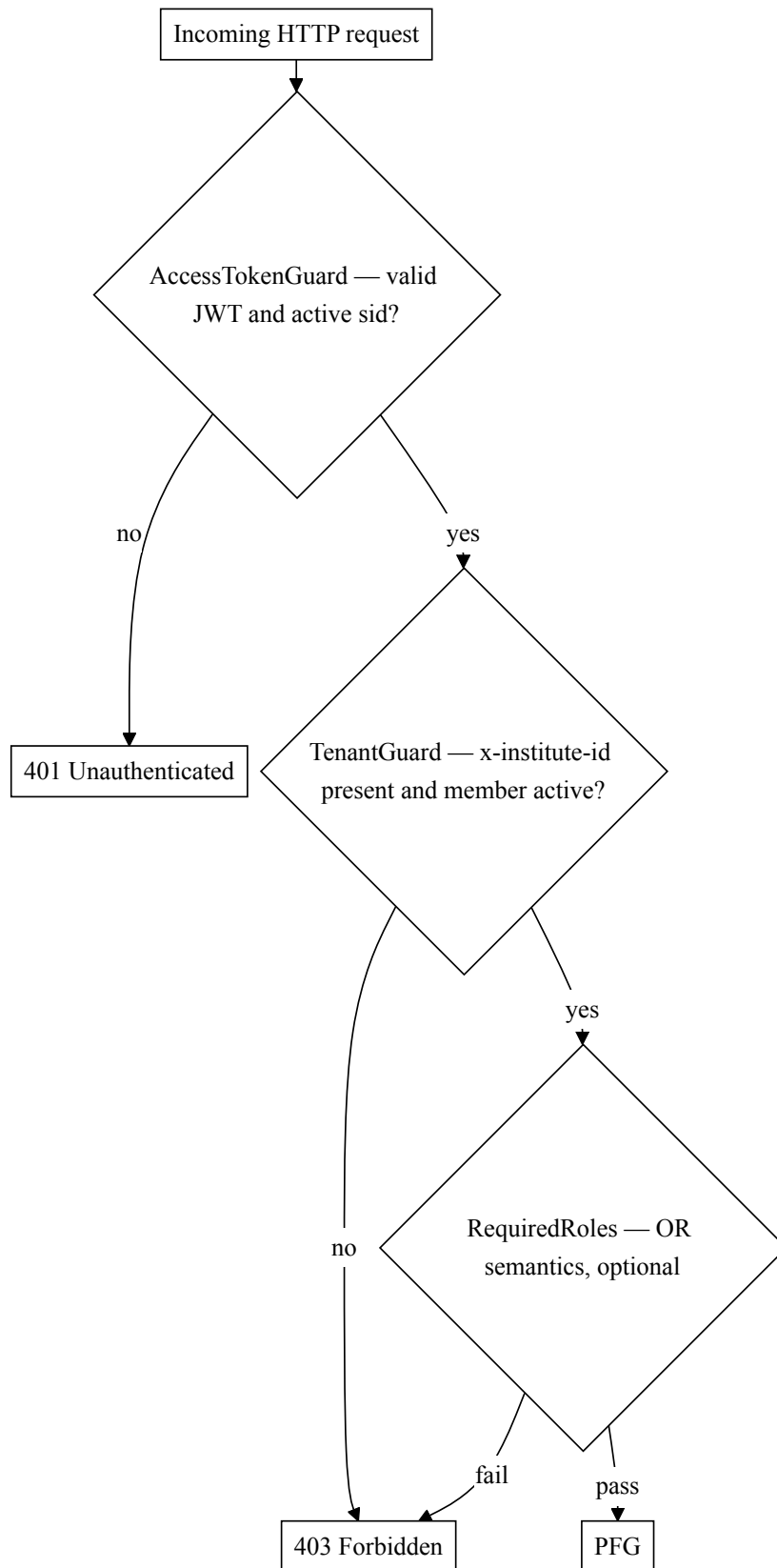


Figure 7.4. Authorization decision chain for a tenant-scoped request, part 1: the guard-stage decisions.

CatLium EduTech — Authorization Decision Chain (permission + scope)

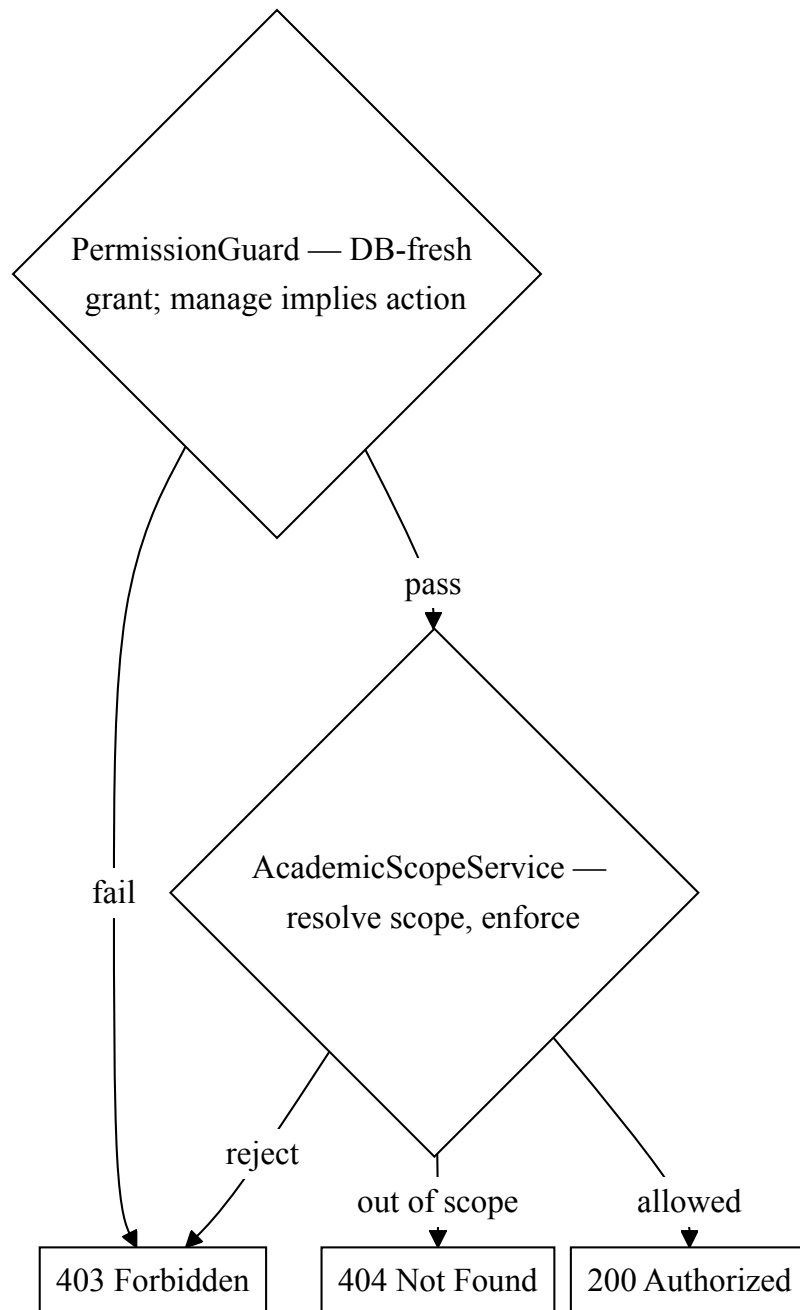


Figure 7.5. Authorization decision chain for a tenant-scoped request, part 2: permission evaluation and academic-scope enforcement outcomes.

CatLium EduTech — Academic Scope Resolution and Enforcement

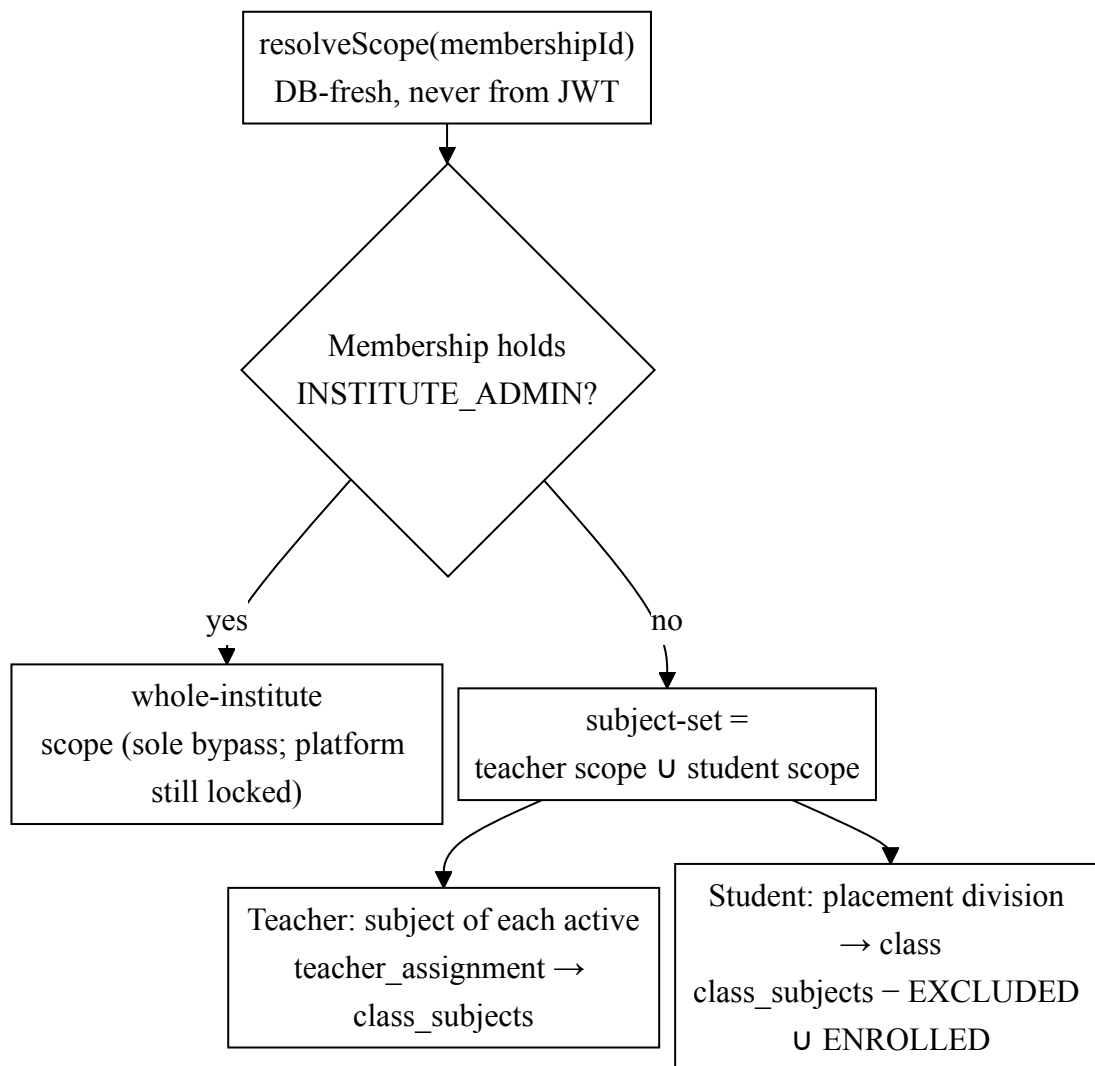


Figure 7.6. Academic scope — resolution of the actor's subject set from database state.

CatLium EduTech — Academic Scope Enforcement on Requests

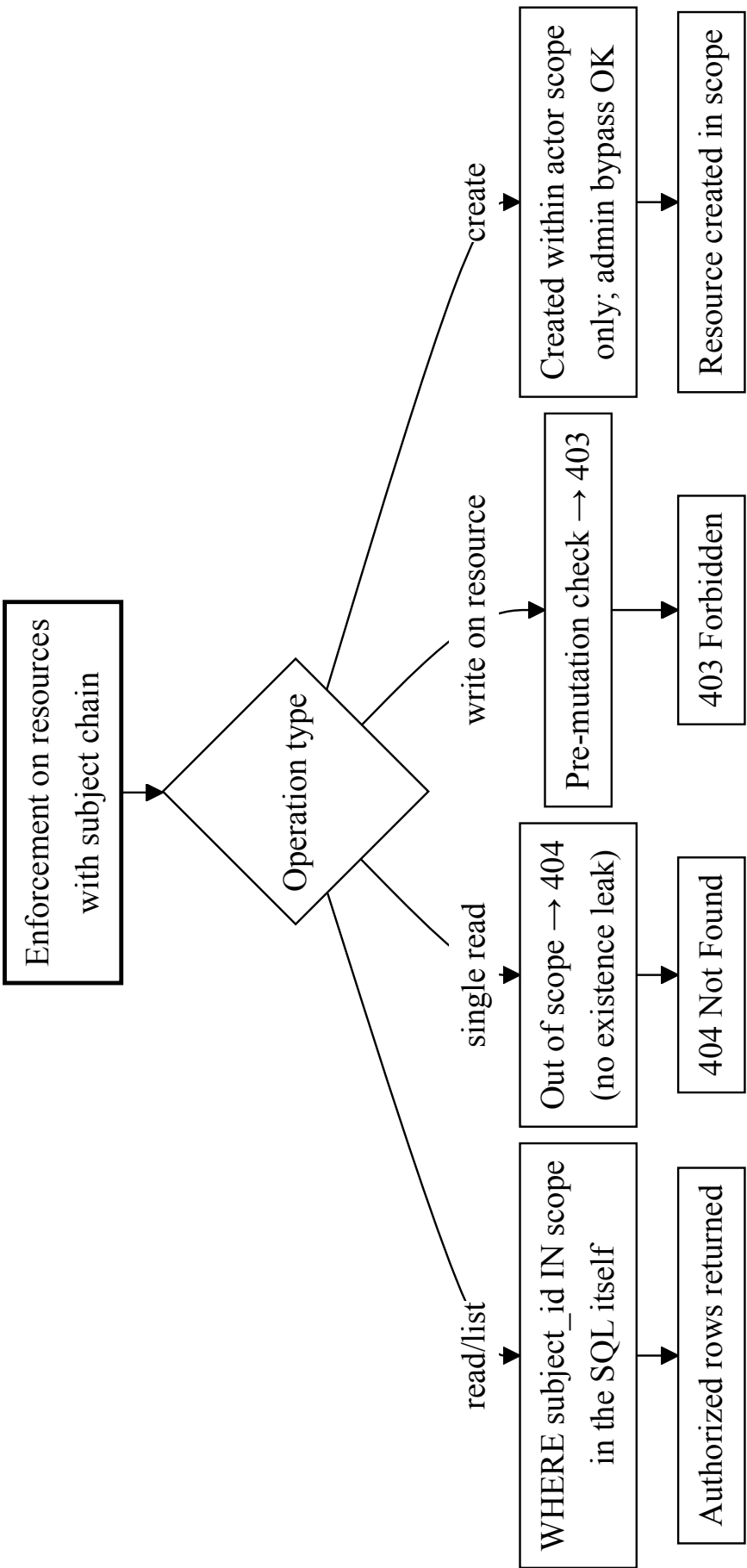


Figure 7.7. Academic scope — enforcement outcomes (reads 404, writes 403).

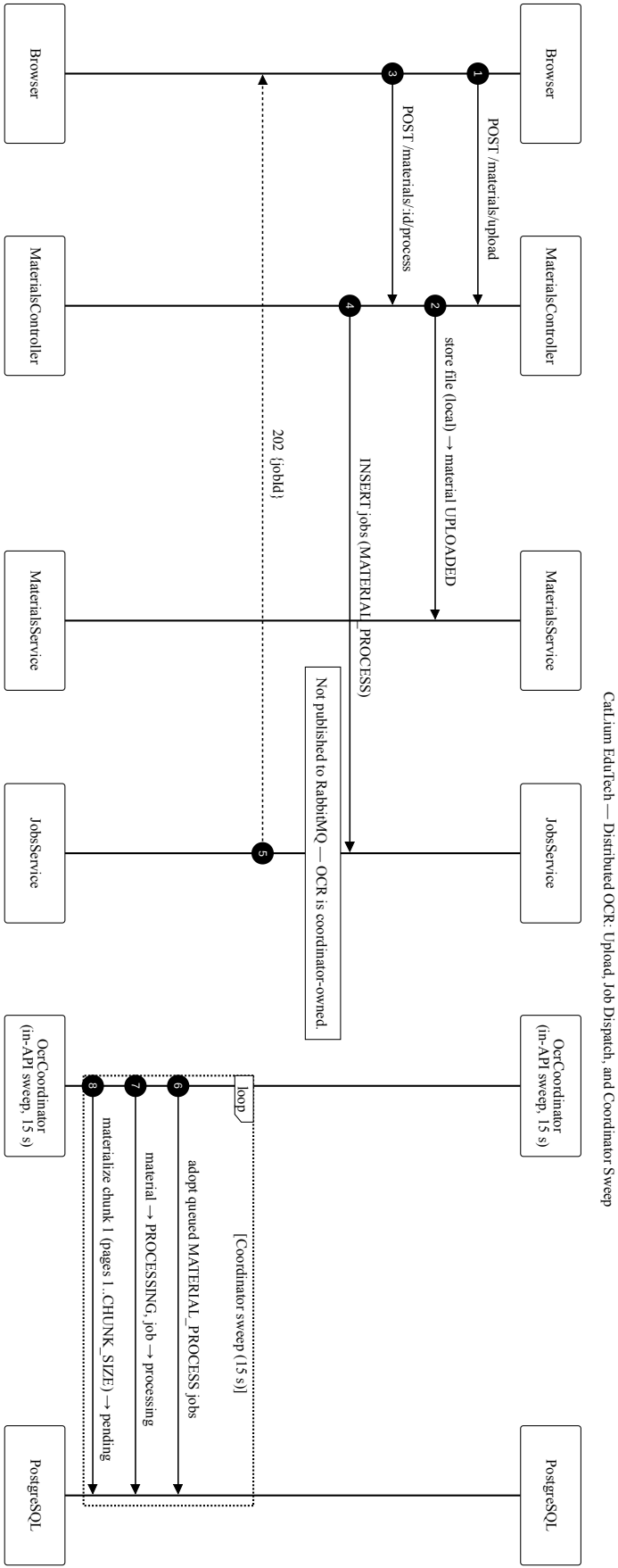


Figure 8.1. Distributed OCR material processing, part 1: upload, job dispatch, and the coordinator sweep.

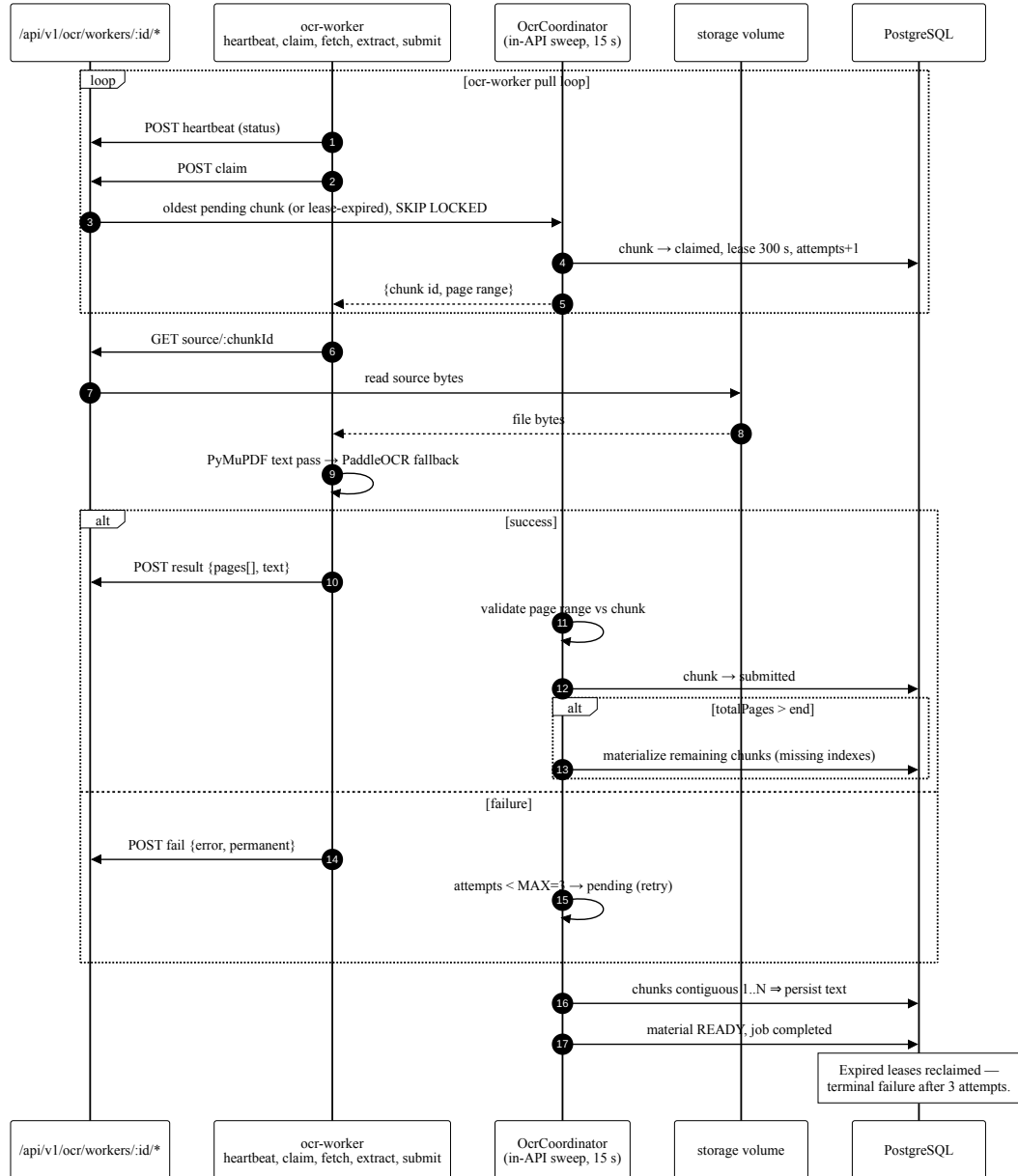


Figure 8.2. Distributed OCR material processing, part 2: worker pull, extraction, and settlement.

CarlIum EduTech — Async job state lifecycle

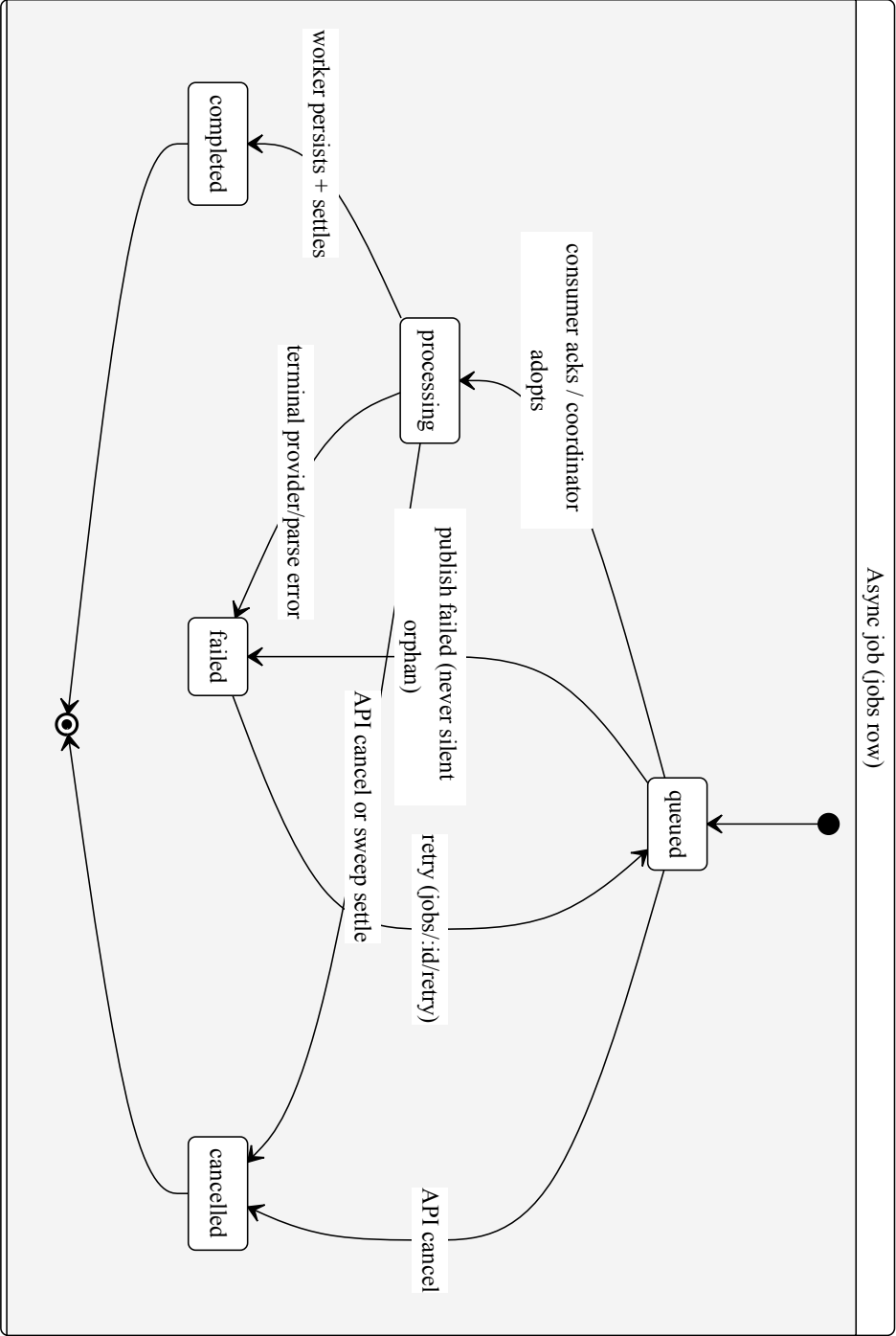


Figure 8.3. Job state lifecycle.

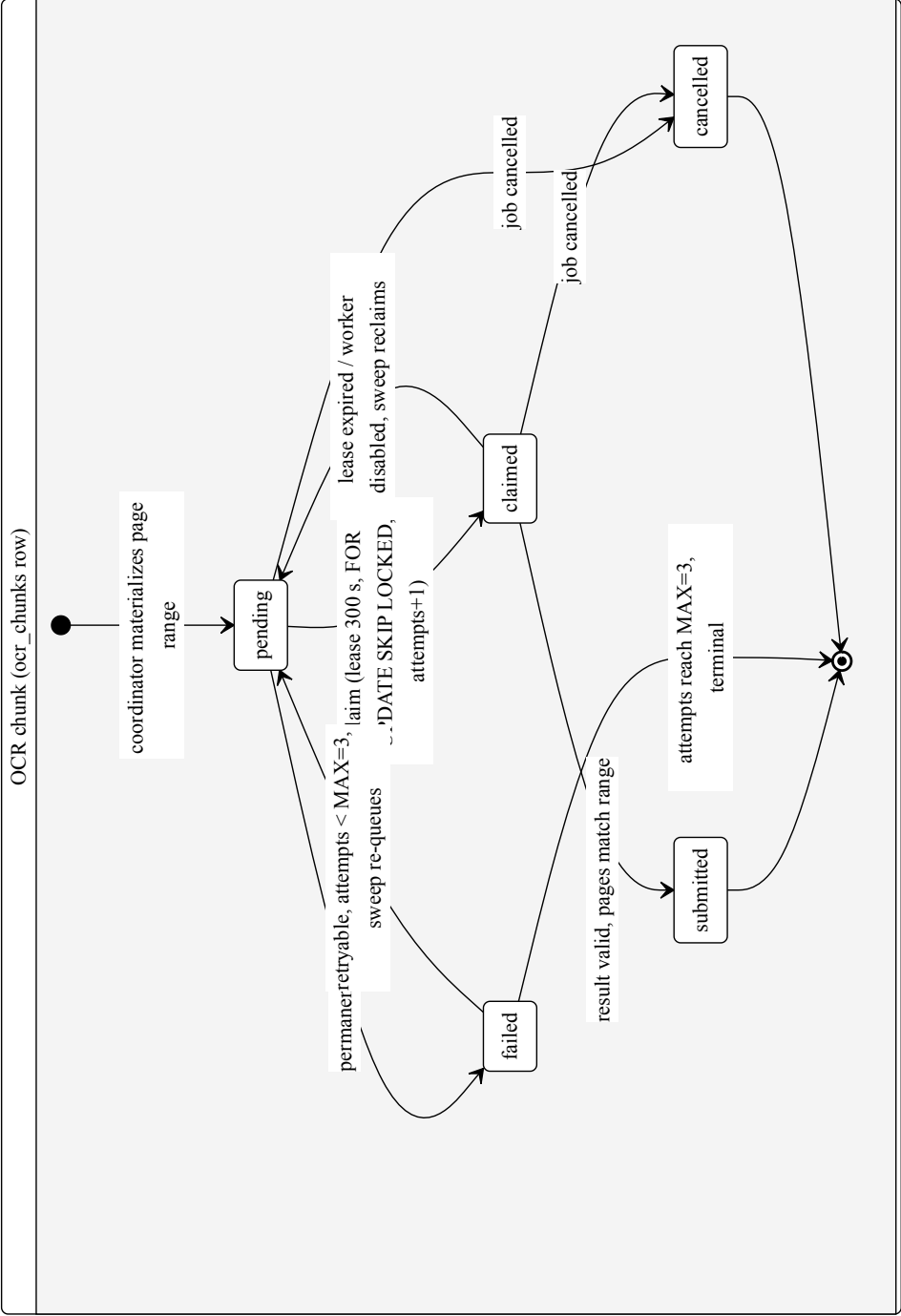


Figure 8.4. OCR chunk state lifecycle.

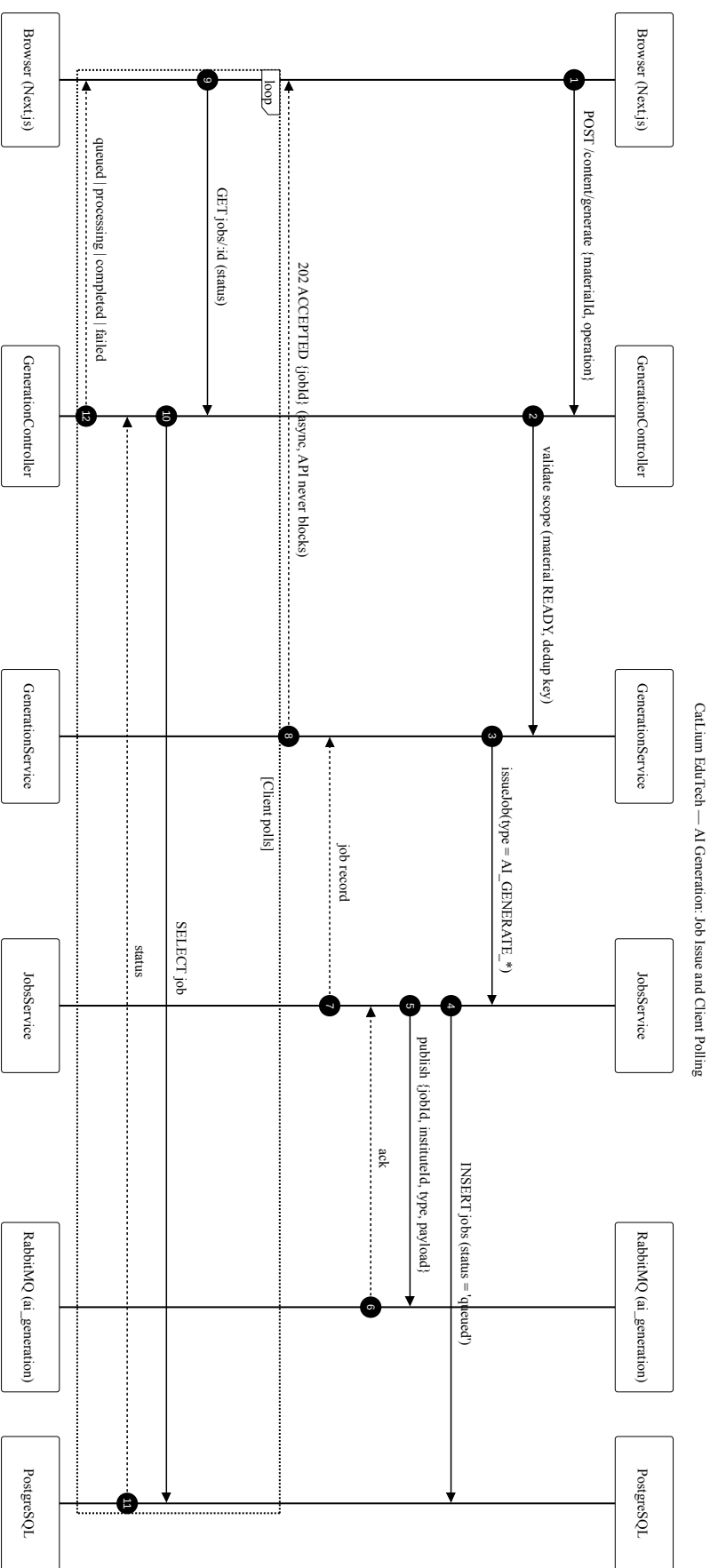


Figure 8.5. Asynchronous AI generation, part 1: job issue and client polling.

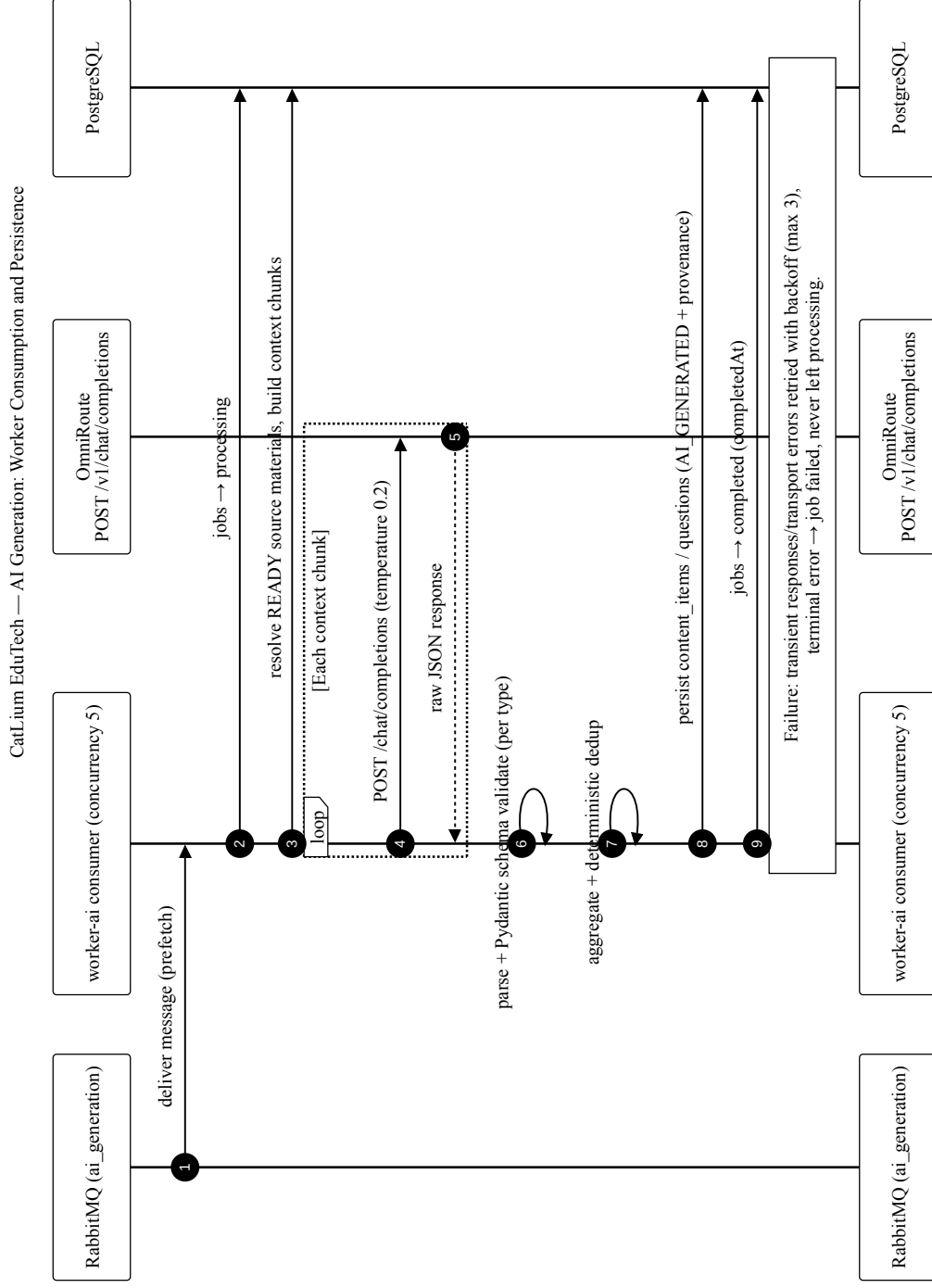


Figure 8.6. Asynchronous AI generation, part 2: worker consumption, OmniRoute calls, and persistence.